

ELEKTRONIK &
TECHN.
INFORMATIK

EMR-Mapping

Diploma Thesis

HTL BULME Graz-Gösting

department for

Elektronik & technische Informatik

submitted by

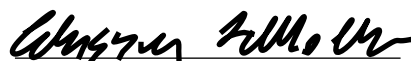
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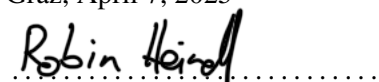


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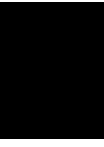
Last but not least, we would also like to thank Markus Kolli for assisting in building the case for the device.

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Project definition

1.1 Goal

Electromagnetic radiation (EMR) is everywhere, and nobody can escape it. Since technology all over the world gets more advanced by day, the EMR pollution will only increase. This makes people wonder if it could cause serious harm to humans. From this fear a lot of conspiracy theories arise.

To clear things up and answer the question, if the increase in EMR is damaging to us, we want to create a device, which should be capable to measure the electromagnetic radiation in our environment. Additionally, the device should also be able to save the data along with the GPS location. That way we are able to create a map with the EMR in our environment. In addition, we will research scientific studies to better understand the effects of increasing electromagnetic radiation on our environment.

1.2 Product

1.2.1 Hardware

Lisa Kolli

The hardware part of the project should be able to measure the electromagnetic radiation. This device should store all data on a storage device, including the GPS coordinates, of the current location. The data should also be displayed in real time, on a display. Since the device needs to be portable, it should be powered by a transportable power source.

1.2.2 Software

Robin Heindl

The software part should be able to save, read and process the collected data. Additionally we also want to show the data on a map.

1.2.3 Literature Research

Letissa Adlboller

Since data without information is almost useless, this project contains a literature research part. It should inform about possible health risks which could accrue, from electromagnetic radiation.

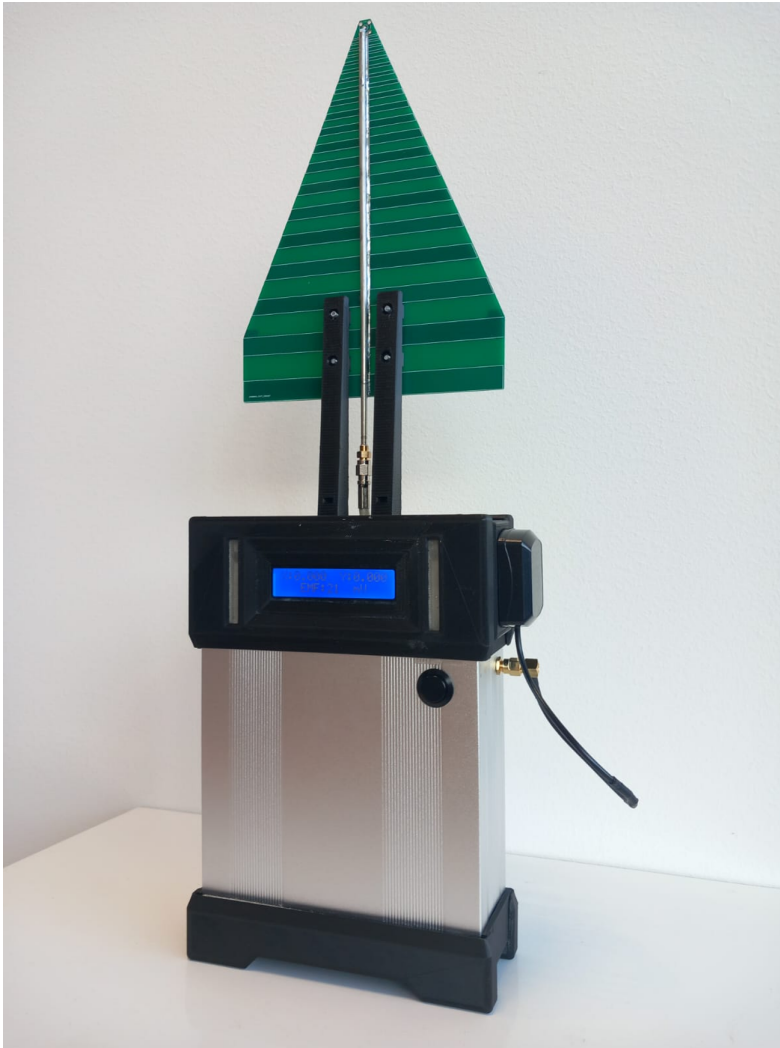


Figure 1.1:
Finished device

CHAPTER 2

Topics

2.1 Electromagnetic radiation

Lisa Kolli

What is electromagnetic radiation?

Electromagnetic radiation, in short EMR, is a set of waves which distributes through space, without a physical conductor. It consists of oscillating electric and magnetic fields that travel at the speed of light.

[1][2][3]

2.2 Location of EMR

Robin Heindl

Electromagnetic radiation (EMR) is everywhere. It comes from natural sources like the sun, which gives us visible light and heat, and from the earth itself, which emits infrared energy. At the same time, human-made devices, such as radio transmitters, mobile phones, and microwave ovens also produce EMR. In essence, whether it's the light that illuminates our world or the radio waves that enable communication, EMR fills our environment all around us.

It also plays a crucial role in communication, medicine, and various technologies. [4] [5]

2.3 Ionizing and non-ionizing radiation

Letissa Adlboller

Ionizing Radiation

Ionizing radiation is the energy released by atoms which travel in the form of electromagnetic waves (gamma or X-rays) or particles (neutrons, beta or alpha). This type of radiation has enough power to remove electrons from atoms and can therefore change the chemical composition of the material it interacts with.

Ionizing radiation can come from natural sources such as cosmic rays as well as from naturally occurring radioactive materials which can be found in soil, water and air. The exposure can

also come from man-made sources the most common sources are medical devices, including Computer Tomography (CT) scanners, X-rays or Fluoroscopy.

When ionizing radiation interferes with living tissue it can damage the DNA either resulting into cell death or mutation (cancer). High dose exposure can cause acute effects such as skin redness, hair loss, radiation burns, or acute radiation syndrome and even impair the functioning of organs [6] [7].

Non-Ionizing Radiation

Non-Ionizing radiation is also a type of electromagnetic radiation. Other than ionizing radiation, non-ionizing radiation does not have enough energy to detach electrons from atoms.

Humans are exposed to both natural-sources as well as man-made-sources of non-ionizing radiation. Natural sources include the earth's natural electric and magnetic fields, the light and heat from the sun as well as lightning. Artificial sources include wireless technologies such as mobile phone base stations, mobile phones, radio and TV-broadcast antennas as well as power lines, microwave ovens and much more.

Direct and intense contact with non-ionizing electromagnetic radiation can cause heating of the tissue, but is not proven to cause other health problems [8][9].

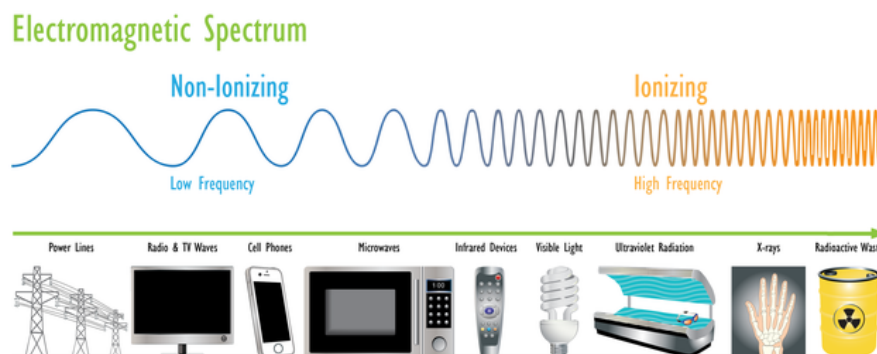


Figure 2.1:
Electromagnetic
spectrum [10]

CHAPTER 3

Specifications

3.1 Software

Robin Heindl

3.1.1 GPS integration

- Retrieves latitude and longitude coordinates
- Updates values continuously when available
- Saves GPS and EMR data synchronously

3.1.2 Data storage

- Saves readings to CSV file
- Uses user-controlled button to start/stop saving data
- CSV format: Longitude, Latitude, EMR-value

3.1.3 Visualization & analysis

- Map generation to see collected data in Graz
- Statistical calculations (mean, max, min, standard deviation)
- Graphical analysis (bar chart, line chart, boxplot)

3.1.4 HTML report

- Automatically generates an interactive HTML page
- Includes map & statistical graphs
- Deployable to school's web hosting system

3.2 Hardware

Lisa Kolli

3.2.1 Reading time

- 1 time per second

3.2.2 Frequency range

- From 800 MHz up to 6 GHz

3.2.3 Component specifications

Storage medium

- Removable and portable
- Minimum 2 GB
- User should be able to decide if data will be stored

Power supply

- Portable
- Minimum 3 hour battery life

Display

- Display current location
- Display EMR-value with at least two decimal places

3.2.4 Case

- Portable
- IP-class: 43
- Maximum 1 kg heavy
- Maximum 300 mm high and 200 mm wide

3.3 Literature Research

Letissa Adlboller

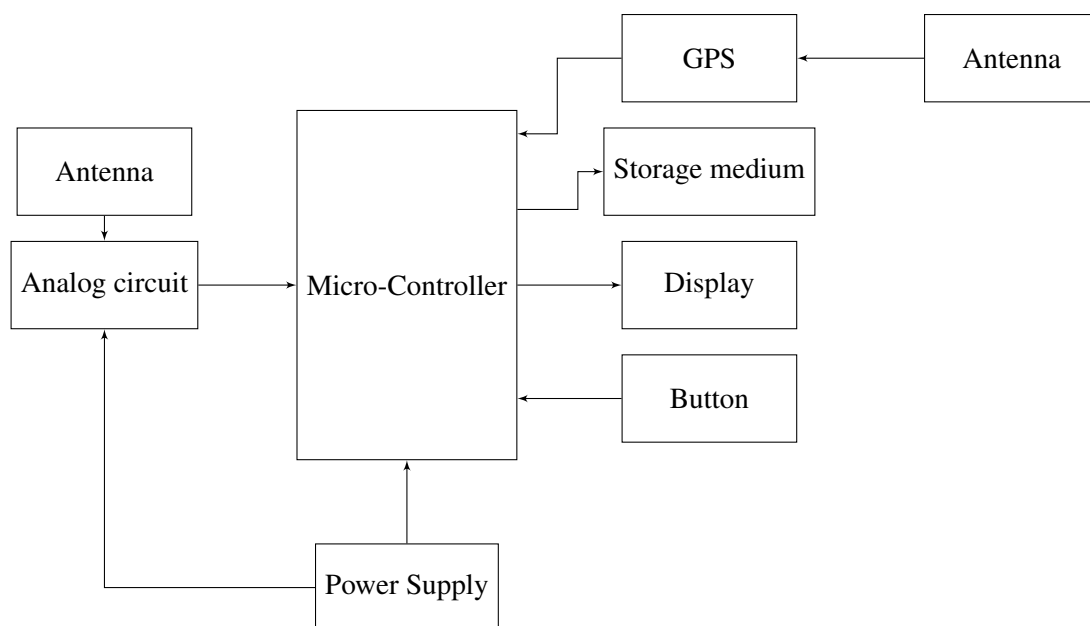
- Various study types
- Studies from 1995 to 2025

Concept

4.1 Circuit

Lisa Kolli

4.1.1 First draft



4.1.2 Microcontroller

Raspberry Pie

There are a lot of different microcontrollers under the Raspberry pie name. Some well-known examples include the Raspberry Pie Pico and the Raspberry Pie 5.

The Raspberry Pie series, which include Raspberry Pie 1 to Raspberry Pie 5, are microcontrollers with the power of a cheap laptop. It is even possible to install a user interface on these kinds of microcontrollers. Due to their high performance, the Raspberry Pie boards are more expensive than other micro controller boards. Additionally, these boards are an overkill, since this project doesn't require that much performance.

On the contrary the Raspberry Pie Pico series is more similar to other microcontrollers on the current market. It has significantly lower performance than other Raspberry Pi boards, which would make it more fitting for this project.

Arduino

When someone mentions microcontroller, the first board that comes to mind is probably one of the Arduino boards. The Arduino series includes boards like the Arduino Uno, Arduino Mega and Arduino Nano. Because of the wide variety, there is a microcontroller board for everyone and everything. However because of the brand's popularity, Arduino microcontroller boards are more expensive than other microcontroller boards.

ESP32

Another well-known microcontroller board is the ESP32. It is similar to the Arduino boards, though it costs a lot less than the Arduino micro controller boards. Another advantage of the ESP32 is that it includes a Wi-fi module. However this project is not in need of an Wi-fi module.

Conclusion

For this project, we need a simple and affordable microcontroller that is easy to use. Since we already have an ESP32, it makes sense to use this type of microcontroller.

Because of these factors, the micro controller in the project will be an ESP32 micro controller board.

4.1.3 Power Supply

Mains power

One of the probably easiest way to supply the product with energy is through mains power. Most smartphone chargers provide the exact power needed to supply an ESP32 board. Nevertheless, we would also deal with a lot of issues, if we decide to supply the project with mains power.

The cause of the problems would be the pollution and interferences of the mains power. Since the task of the project is to measure electromagnetic radiations, it would impractical to get pollution in our system through the power supply. In the worst case, it could distort the values we measure. To mitigate this, a 50 Hz notch filter could be used to reduce interference. Even if that would

work, supplying the device with the mains power supply would not be a solution, since the device needs to be portable and the mains power supply is tethered.

Rechargeable battery

To solve the issue with the portability, a rechargeable battery would be a solution. Another thing that would not occur with a rechargeable battery is the pollution and interferences through the power supply, since a battery has almost no electromagnetic radiation.

However, rechargeable batteries are not always small or cheap. Many of them take up a lot of space, which could be problematic since the device needs to be compact and portable. Additional rechargeable batteries can be pretty expensive.

That being said, using an already existing power bank, would be a really easy solution, since it is easy to recharge, and it supplies the device with the perfect amount of power.

Primary battery

Another way to solve the issue with the portability, would be primary batteries. Primary batteries have the advantage, that they, like rechargeable batteries, produce almost no electromagnetic radiation. They are also pretty cheap and easily available.

This is also one of the main disadvantages. Primary batteries are cheap, because they are not reusable. After they are drained, the batteries need to be discarded. This causes a lot of unnecessary waste. In addition, you also need to buy new batteries whenever the batteries are drained.

Since batteries can be used in pairs, there is a possibility to create every kind of power supply, no matter for what device. This also includes this project.

Conclusion

The power supply will be a rechargeable battery, to be specific a power bank. One of the reasons for that decision is the fact, that there is no need to dispose a rechargeable battery when they are drained, since they can easily be recharged. This is better, since there is no need to buy a new battery, which is good for us and the environment.

4.1.4 Antenna

Wire antennas

Wire antennas are the most commonly used antennas, since they are easy to install. A wire antenna is like the name already says, an antenna made out of a most of the times long wire, which is suspended over the ground. Also important to know is that the length of the wire has no relation to the wavelength of the signal.

In general wire antennas are subdivided in different types, such as dipole antenna, short dipole antenna, half-wave dipole antenna, folded antenna, loop antenna, helical antenna and so on.

The dipole antenna is the most significant form of radio frequency antenna. The antenna length

has to be at least half of the maximum wavelength to function properly. The vast majority of wire antennas are dipole antennas, which come in various types.

One example for a dipole antenna would be the short dipole antenna, who is not called that way because of the antenna size. Moreover the antenna size does refer to the relative wavelength of the signal. The short dipole antenna can send and receive frequencies between 3 kHz and 30 MHz.

There is also a folded dipole antenna, which is mainly used as a TV receiver. It consists of two conductors, who are connected with each other. The folded dipole antenna has a frequency range from 3 kHz to 300 GHz.

Even though already two dipole antennas got explained, there are a lot more types, for instance the half wave dipole antenna, also known as Hertz antenna. The half wave dipole antenna operates in a frequency range between 3 kHz and 300 GHz.

Then there is also the loop antenna, another type of the wire antenna. The loop antenna consists of a wire, which is creating one or multiple loops. This kind of antenna is highly efficient in receiving frequencies between 300 MHz and 3 GHz, since it needs less electricity compared to other antennas. However, the performance of a loop antenna is dependent on its design and placement.

For any kind of device, the wire antennas would come in handy, since they are easy to install and most of the times, small and light weight. Additionally, most of the wire antennas are easily available. However, the frequency range for most of the wire antennas is not big enough.

[11][12]

Log periodic antennas

One of the probably most practical types of antennas, for this project, are the log periodic antennas, considering the wide range of frequencies they can cover. The majority of log periodic antennas can receive and send frequencies between 30 MHz and 6 GHz, however the range completely depends on the antenna layout.

A typical log periodic antenna consists of various half-wave dipole antennas, who increase in length.

Log periodic antennas are primarily used for TV signal reception and for signal monitoring applications.

An advantage of this antenna type is, that the antenna has a compact structure, even though it has a wide frequency range. That being said, one of the main downsides of log periodic antennas is, that they are quite expensive.

[13][12]

Aperture antennas

Aperture antennas are antennas, who have an aperture at the end. The antenna comes in various sizes and forms, it can be circular, rectangular, square or cone. The most common kind is the circular aperture antenna, since it is a normal satellite dish. Depending on the type of aperture

antenna, a different frequency range can be gained and send.

The waveguide antenna, which is a form of aperture antenna, is capable of working with a frequency range from 300 MHz to 300 GHz.

Another aperture antenna type would be the slot antenna, this kind of antenna is usually a flat metal plate, with one or more holes or slots cut out of it. Usually, usage of this antenna is between 300 MHz and 24 GHz.

Then there is also the horn antenna, which works in a frequency range from 300 GHz to 30 GHz. The horn antenna is waveguide shaped like a horn.

One of the main reasons why these antennas are used, is because of their great performance and the high frequency range. However, the aperture antennas can be quite spacious and expensive. [14][12]

Microstrip antennas

Microstrip antennas, also known as patch antennas, are usually fabricated by using photolithographic techniques on a printed circuit board. It comes as different types of antennas, a microstrip antenna can be a dipole antenna, a slot antenna and a lot more.

It is really small and lightweight, as well as very cost-effective. But the microstrip antenna has a low efficiency and is quite sensitive to environmental factors, like temperature and humidity. [15][12]

Array antennas

An array antenna, often also called phased antenna consists of at least two antennas of a different antenna type, for instance three dipole antennas. Most of the time the antennas are sorted in rows and columns.

The signal of the individual antennas are combined, because of that the array antennas have an improved performance. However, they cost a lot and are also really complex. Additionally array antennas can be more vulnerable to environmental factor, like rain and wind.

[16][12]

Conclusion

After thoughtfully thinking about the options, the decision is, that a log periodic antenna would make the most sense, when it comes to receiving electromagnetic radiation. Reason for that is the wide frequency range this kind of antenna can obtain.

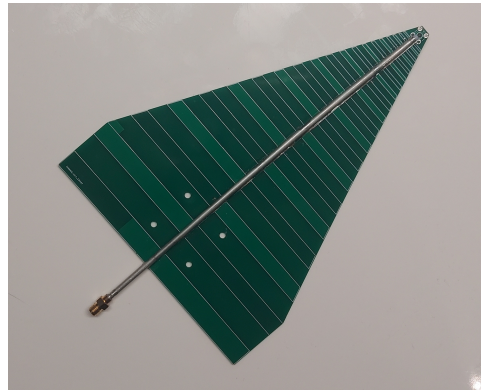


Figure 4.1:
Log periodic
antenna

4.1.5 GPS

GPS Data via WiFi-triangulation

Wi-Fi-triangulation uses available access points in the area, to determine the exact location of the device. As it says in the name of the method, it needs a connection to at least three access points to function.

Given that these kinds of connections are unlikely outdoors, which is also where the measurements will be taken, this method is not practical for this project.

GPS module

The by far easiest solution would be to use an already for the microcontroller existing module. These kinds of GPS modules are not only easy to use, furthermore they are cheap.

On the downside they tend to be quite slow when it comes to finding the first position.

GPS data via smartphone

One rather complicated way to get the current location, is with the help of a smartphone. This method works in cooperation with the ESP32 Wi-fi Module, which is used to build a connection between a microcontroller and a smartphone. Then the smartphone sends its GPS data to the microcontroller.

Since there would be a continual connection between the microcontroller and a smartphone, the electromagnetic pollution could be a problem. In the worst case it could distort the measured values.

Conclusion

Because of the simplicity and availability the GPS module will be used for this project.

4.1.6 Storage medium

Online storage medium

Since this project contains an ESP32, there is a possibility to store the data, with the help of an internet connection. This leads to a lot of different possibilities, for example cloud storage or wireless data transfer.

However using any kind internet connection, causes electromagnetic pollution and this would distort the values we measure. Which would not be ideal for this project.

USB Stick

On the contrary to saving the data with the help of Wi-fi, there would be little to no electromagnetic pollution. This kind of storage medium would also be removable, which would be ideal for that project.

SD Card

Another storage medium which can be used with almost no electromagnetic pollution, would be the SD Card. It is not only removable but also really small, because of that the medium would be more convenient, since it doesn't take up much space.

Conclusion

For this project we chose to use the SD Card, since it is easy available and removable.

4.1.7 Display

LC Display

Liquid Crystal Displays are, especially in the micro controller area, an often used way to show data. They are easy to use, since there are multiple libraries to help control the display.

Instead of emitting light, LC displays modulate or block light, which is one of the main reasons for their energy efficiency. To produce an image it uses black light.

LC Displays have a parallel interface, which consists of 16 pins.

One of the downsides is that LCDs are not quite presentable, considering they only have two colours, black and white. Even so they are easily available, which makes them a good choice after all.

OLED Display

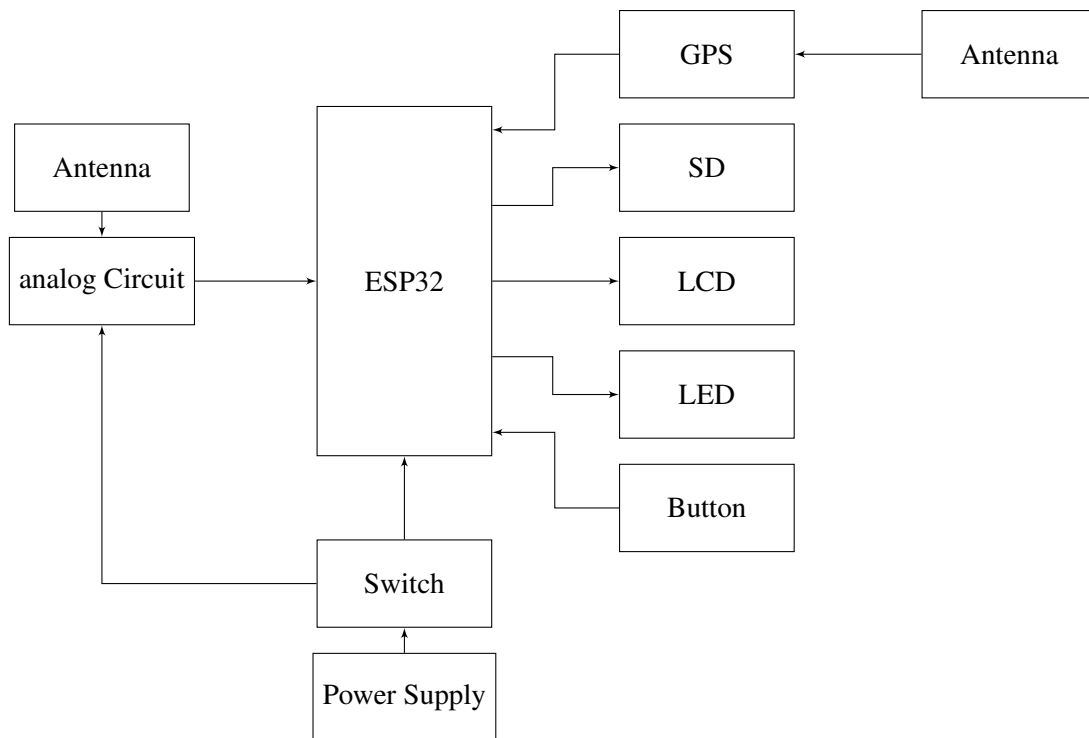
Instead of LCDs, Organic Light Emitting Diodes Displays do not need a white backlight source. This makes them thinner and even more efficient. Additionally the image quality of OLED Displays is also better. Not only can it display multiple colours it has smaller pixels as well, which makes the display more presentable.

However, because of the advantages it costs more than other displays. Due to the price it begs the question if the device really needs a higher resolution display.

Conclusion

Even though OLED Displays have more advantages than LC Displays, it is not necessary to invest more money into the device, since we only need to display numbers. Therefore LCD meets our requirements and will be used in the device.

4.1.8 Block diagram



4.2 Case

Lisa Kolli

4.2.1 Material

Polylactic Acid

Polylactic Acid or better known as PLA, is probably the most popular material, which is used for 3D printing. It is rather cheap and easily available. In addition it is quite stiff and has a lot of strength. However it is not suitable outdoors, since the sunlight can make the material brittle, as well as bleach the printed object.

Another benefit of PLA is, that there is a biodegradable version of it, which would make the product considerably more environmental friendly.

To make use of PLA, we would need a 3D printer, fortunately we already have one. The printer must be capable of heating up the bed temperature between 45°C and 60°C as well as heating

up the extruder between the temperature 190°C and 220°C.
[17]

Acrylic Styrene Acrylonitrile

The short form of Acrylic Styrene Acrylonitrile and also more known name is ASA. ASA is similar to PLA, it is also a material which is used in 3D printing. Unlike PLA, ASA is much more resistant to UV rays, making it much more suitable for outdoor use. However these benefits are part of the reason, why the material costs far more. In addition, the material should only be used as a printing material in non closed rooms, since it can produce dangerous fumes.

Unfortunately there is no biodegradable version of Acrylic Styrene Acrylonitrile, causing the material to be considerably less environmental friendly. Besides that the 3D printer needs to be capable of heating up to way higher temperatures, than with PLA. The bed must have a temperature between 90°C and 110°C and the extruder must heat up between 220°C and 245°C.
[18]

Wood

Another good material would be wood, which is not only aesthetically pleasing, but also in every hardware store available. In addition it is highly environmental friendly, given that new trees can always be planted. A known characteristic of wood is, that the electric conductivity is really low. For that reason there are many benefits and disadvantages in using wood as a case material.

An advantage, for example, is that thanks to the low electric conductivity, it is almost impossible to create a short circuit through the housing. Additionally wood is a natural insulating material, in other words it does absorb and block electromagnetic radiation. Even though this characteristic sounds quite handy, for this to work, the housing must be build with solid wood.

This kind of wood is quite heavy and tends to be more expensive. Besides that the amount of work to make a case out of wood is very high and to make the whole thing work out, a lot of knowledge about wood work would be needed.

Aluminium

One of the probably most used materials for cases in the electronic field is aluminium. Mainly due to its lightweight, high strength and especially its excellent thermal conductivity. As a result aluminium prevents overheating, since it can transfer heat away from electronic components. Furthermore aluminium is corrosion resistant, due to the fact that it naturally forms a thin oxide layer on it's surface. Therefore it is also usable outdoors and in other harsh environments.

However aluminium is, similar to wood, rather complicated to work with. Considering that special machinery and a wide range of knowledge is needed to work with aluminium.

Even though there are a lot of advantages in aluminium's electrical conductivity, a major disadvantage is that there is a danger of creating a short circuit though the housing, which would be fatal to the project. Besides that aluminium tends to get quite cold outdoors, what makes it uncomfortable to carry around.

[19]

Conclusion

To create an ideal case, the materials polylactic acid and aluminium will be used for the housing. Not only is the aluminium perfect to protect the electronics, it also shields the circuit from electromagnetic pollution. To make working with the aluminium easier, the housing will be build with the help of 3D printed elements, made of PLA. This also helps to make the case look much more appealing.

4.3 ESP32 Firmware Software

Robin Heindl

4.3.1 Programming Language

The software for ESP32 is implemented in C/C++. These languages are widely used for embedded systems due to their efficiency, hardware control, and compatibility with the ESP32 microcontroller.

4.3.2 GPS - Library Options

TinyGPS++

TinyGPS++ is a popular and well-maintained library for handling GPS data on embedded systems. It is designed to efficiently parse NMEA sentences, which are the standard format for GPS communication. One of its key advantages is its low memory footprint, making it ideal for microcontrollers like the ESP32 that have limited RAM. Additionally, TinyGPS++ offers built-in functions to extract useful information such as latitude, longitude, altitude, and speed. It is also capable of handling satellite signal quality, which can help improve the accuracy of location tracking. Due to its optimized code structure, TinyGPS++ ensures that the ESP32 can process GPS data without excessive CPU load, allowing other tasks to run simultaneously.[20]

NeoGPS

NeoGPS is known for its high precision and efficiency in processing GPS data. Unlike TinyGPS++, NeoGPS is designed to be more customizable, offering developers the ability to tweak various settings to suit specific needs. One of its most significant advantages is that it processes GPS data in a structured manner, reducing memory usage by only keeping necessary data. This can be particularly useful when working with large amounts of GPS information. However, the increased customization comes at a cost. NeoGPS requires more setup and configuration compared to TinyGPS++, making it slightly more complex to use. Despite its learning curve, it is a solid choice for applications that require high accuracy and performance optimization.[21]

Adafruit GPS Library

The Adafruit GPS Library is a beginner-friendly option developed by Adafruit Industries, specifically designed for use with their GPS modules. It provides a simple interface to communicate with GPS devices, making it easy to set up and retrieve location data. One of its benefits is that it has predefined examples and tutorials, which can help new developers integrate GPS functionality

into their projects quickly. However, the library is less optimized for real-time data processing, meaning it might introduce slight delays when dealing with fast-moving objects. Additionally, compared to TinyGPS++ and NeoGPS, the Adafruit GPS Library has fewer advanced features, making it less suitable for high-precision applications.[22]

Mikal Hart's TinyGPS

Mikal Hart's TinyGPS is the predecessor of TinyGPS++ and was widely used in early GPS projects. It is a lightweight and straightforward library that provides the basic functionalities needed to decode NMEA sentences and extract location data. Due to its minimalistic design, it requires very little memory, making it suitable for microcontrollers with even lower RAM than the ESP32. However, compared to TinyGPS++, it lacks several advanced features, such as support for multiple satellite constellations and improved signal processing. This makes it a less attractive choice for modern embedded systems that require higher accuracy and additional metadata extraction.[23]

Conclusion

For this project, TinyGPS++ was chosen due to its balance between efficiency and ease of use. It provides the necessary parsing capabilities while being optimized for resource-constrained environments like the ESP32. TinyGPS++ offers good performance, simple implementation, and a lightweight structure, making it ideal for embedded systems where resources must be used efficiently. While NeoGPS offers better precision, its added complexity was unnecessary for this application. The Adafruit GPS Library, although easy to use, lacks real-time optimization, and Mikal Hart's TinyGPS does not offer the advanced features needed for reliable GPS data handling. Therefore, TinyGPS++ was the best choice to ensure smooth and reliable GPS data processing on the ESP32.

4.3.3 LCD-Display - Library Options

LiquidCrystal_I2C.h

The LiquidCrystal_I2C.h library is a popular and efficient choice for controlling I2C-based LCD displays, particularly in embedded systems with limited resources such as the ESP32. It provides an easy-to-use interface to display text on character-based LCDs while keeping memory usage low. One of the main advantages of this library is its ability to control multiple LCDs using only two wires (I2C protocol), which significantly reduces the number of pins required for display operation. This library supports a variety of character-based displays, making it versatile for many applications. Its memory-efficient nature makes it an excellent choice for resource-constrained environments like the ESP32. It is ideal for projects where simplicity and minimal resource consumption are priorities, such as displaying numerical data like EMR values.[24]

U8g2

The U8g2 library offers support for a wide range of graphic LCDs, OLED displays, and other monochrome displays. It includes high-quality fonts and better graphics support compared to LiquidCrystal_I2C.h, allowing developers to display more complex visual content, such as custom images and detailed graphics. U8g2 provides a more extensive set of features, including multiple display drivers, advanced text rendering capabilities, and support for various display sizes and orientations. However, its richer feature set comes at the cost of higher memory consumption. This library can be a good choice for applications that require high-quality graphical content but may not be ideal for applications with tight memory constraints, such as those using the ESP32 with limited resources for other tasks. [25]

Adafruit GFX & Adafruit SSD1306

The Adafruit GFX library, paired with the Adafruit SSD1306 library, is specifically designed for controlling OLED displays, particularly those based on the SSD1306 driver. This combination offers high-resolution graphics and a wide range of features, making it ideal for more sophisticated displays. The Adafruit GFX library provides functions for drawing shapes, displaying text, and even rendering custom graphics, allowing for rich, visually appealing user interfaces. The Adafruit SSD1306 library is optimized for OLED displays with excellent contrast and sharpness. While this combination is excellent for high-resolution displays and graphic content, it tends to be heavier on memory usage compared to simpler libraries like LiquidCrystal_I2C.h. Thus, it is best suited for projects that require detailed visuals or OLED-based displays but might be overkill for simpler tasks like numerical data presentation. [26] [27]

TFT_eSPI

The TFT_eSPI library is designed for controlling TFT LCD screens, and it is optimized for ESP32 platforms, offering high performance for graphics and touch-based applications. This library supports a wide range of TFT displays, offering features like high-resolution color displays and touch support. However, for projects like this one that primarily focus on displaying numerical data, TFT_eSPI would be excessive due to its focus on color and touch interfaces. While it provides excellent performance for graphics-heavy applications, its advanced features and higher resource demands make it unnecessary for simple display tasks that do not require such high visual fidelity. This library is more suitable for projects with more complex visual requirements, such as interactive interfaces and graphical displays.[28]

Conclusion

For this project, the LiquidCrystal_I2C.h library was chosen due to its simplicity, lightweight nature, and suitability for displaying numerical data such as EMR values. It is memory-efficient, making it ideal for resource-constrained environments like the ESP32. The U8g2 library, while offering superior graphical capabilities, was not necessary for this project's requirements, as it consumes more memory than required. The Adafruit GFX and Adafruit SSD1306 libraries provide high-quality visuals but are overkill for displaying basic numerical data. The TFT_eSPI

library, while powerful, was unnecessary for this simple application. Therefore, LiquidCrystal_I2C.h was selected as the best option to ensure efficient and straightforward display functionality.

4.3.4 SD-Card - Library Options

SdFat.h

The SdFat.h library provides advanced SD card handling with optimized support for high-speed SPI communication. It is particularly useful for applications that need to handle large amounts of data quickly and efficiently. The library supports high-speed data transfers, allowing for fast reads and writes to SD cards, which is crucial for applications that require quick data storage or retrieval. SdFat.h offers extensive functionality for file management, including the ability to handle large files without running into memory or performance issues. It is a great choice for applications that need reliable and high-performance SD card operations, especially when dealing with large datasets or high-speed data logging.[29]

SD.h

The SD.h library is a simpler and more user-friendly library for working with SD cards. It provides basic functionality for reading from and writing to SD cards, making it an ideal choice for projects that do not require advanced features or high-performance operations. While it is easy to use, SD.h does not have the same optimization for handling large files or high-speed data transfers as SdFat.h. For simple applications that require basic file management on an SD card, SD.h is a suitable choice, but it may not be ideal for projects that demand high efficiency or larger data handling. [30]

FS.h

The FS.h library is the standard file system library used for file management in various storage systems, including SD cards. It provides basic file operations like opening, reading, and writing files. However, it lacks the advanced features and performance optimizations specific to SD card usage. FS.h is a simple and straightforward option for managing files, but it does not offer the same level of optimization or high-speed communication for SD cards that libraries like SdFat.h provide. This makes it more suited for general file system tasks rather than performance-intensive SD card operations.[31]

SPIFFS

SPIFFS (Serial Peripheral Interface Flash File System) is designed for use with the internal flash storage of the ESP32 rather than external SD cards. While it is a great solution for small-scale, internal file storage, it is not designed to work with SD cards. Therefore, SPIFFS is not suitable for projects that require the use of external SD cards. It is best for applications that rely on internal flash memory for data storage, but should be avoided when an SD card is required for removable storage.[32]

Conclusion

For this project, the SdFat.h library was chosen for its superior performance and optimizations for high-speed SPI communication. It is particularly well-suited for handling large amounts of data efficiently, making it the best option for applications that require fast data access or extensive file handling. While SD.h is simpler to use, it lacks the performance capabilities needed for handling large files or complex operations. FS.h is useful for general file management but lacks the advanced features needed for SD card-specific tasks. SPIFFS is not suitable for SD card applications as it is designed for internal flash storage. Therefore, SdFat.h was selected to ensure optimal performance and reliability for SD card operations in this project.

4.4 Python Data Processing and Visualization

Robin Heindl

4.4.1 Programming Language

The software for Mapreader.py is implemented in Python. Python was chosen due to its strong ecosystem for data analysis, visualization, and web-based reporting. Its extensive library support and easy syntax allow for rapid development, making it an ideal choice for processing and visualizing data collected from the ESP32.

4.4.2 CSV Data Processing - Library Options

Pandas

Pandas is a powerful and widely used library for data analysis and manipulation. It provides efficient handling of structured data and built-in support for CSV operations, making it ideal for processing the recorded EMR and GPS data. Pandas allows for quick filtering, grouping, and transformation of datasets, enabling us to extract meaningful insights from the raw data efficiently. Additionally, its ability to work seamlessly with various data visualization libraries enhances the overall analytical workflow.[33]

CSV Module

Python's built-in CSV module provides basic functionality for reading and writing CSV files. While it is lightweight and requires no additional dependencies, it lacks the advanced data manipulation features of Pandas. Using the CSV module would require significantly more manual handling of data structures, making complex transformations difficult. This limitation makes it unsuitable for our needs.[34]

Dask

Dask is designed for handling extremely large datasets by parallelizing operations. While it provides a similar interface to Pandas, it introduces additional complexity due to its distributed nature. Since our dataset remains within manageable limits, Dask would be an overcomplicated solution that offers little benefit over Pandas in this case.[35]

Conclusion

Pandas was chosen due to its versatility, ease of use, and ability to handle numerical and categorical data efficiently. It enables seamless CSV reading and writing while allowing for advanced transformations that facilitate later visualization and analysis.

4.4.3 Heatmap Generation - Library Options

Folium

Folium is an interactive mapping library built on Leaflet.js, allowing for intuitive visualization of geospatial data. One of its greatest advantages is its ability to integrate GPS data seamlessly, making it ideal for plotting recorded EMR values. Folium also supports interactive heatmaps, which can represent large datasets efficiently. This interactivity is particularly useful in our project, as it enables users to zoom, pan, and explore data dynamically.[36]

Matplotlib Basemap

Matplotlib's Basemap extension provides functionality for plotting geospatial data. However, it is mainly used for static maps, which lack the interactivity required for our project. Additionally, Basemap is deprecated, meaning future updates and support are limited.[37]

GeoPandas & Shapely

GeoPandas and Shapely provide advanced geospatial data processing capabilities, enabling spatial analysis and geometric operations. While they are powerful tools, they are not required for our project since our primary objective is visualization rather than complex spatial calculations.[38][39]

Conclusion

Folium was selected due to its ability to generate interactive heatmaps and integrate well with real-world GPS coordinates. A heatmap was chosen over plotting individual data points because displaying thousands of individual points would significantly impact rendering speed and responsiveness. Heatmaps also group similar values together, making it easier to identify hotspots of high EMR levels. Additionally, the color-coded intensity makes it more intuitive to detect areas with extreme EMR values at a glance.

4.4.4 Statistical Analysis & Charts - Library Options

Matplotlib

Matplotlib is a well-established library for creating static visualizations. It allows for highly customizable plots, making it an ideal choice for generating bar charts, line graphs, and boxplots. Since Matplotlib is lightweight and integrates well with Pandas, it is highly efficient for our project.[40]

Seaborn

Seaborn builds on Matplotlib and provides improved aesthetics and advanced statistical visualizations. While it enhances the appearance of charts, its additional dependencies and styling complexity make it unsuitable for our needs.[41]

Plotly

Plotly is a powerful interactive plotting library that allows for zooming and real-time data exploration. However, this interactivity comes at the cost of increased processing overhead and file size, making it less suitable for a lightweight report.[42]

Conclusion

Matplotlib was chosen for its simplicity and extensive documentation, making it ideal for generating statistical plots such as bar charts, line graphs, and boxplots.

4.4.5 HTML Dashboard Generation - Library Options

Branca & Folium

Branca, in combination with Folium, allows us to embed interactive maps directly into an HTML report. This approach ensures that our visualization is portable, requiring no additional software beyond a web browser. By embedding the heatmap into an HTML file, we allow for offline access and easy sharing of results.[43][36]

Dash (Plotly)

Dash provides dynamic web applications with interactive components. However, it requires a dedicated web server, making deployment more complex. Since our project only requires a static report, Dash is not necessary.[44]

Flask/Jinja2

Flask and Jinja2 are excellent for dynamically generating HTML content. However, like Dash, they require a server to function properly, making them unsuitable for our use case.[45][46]

Conclusion

Branca & Folium were chosen for their seamless integration with maps and their ability to generate lightweight HTML reports. An HTML-based report was selected over a web app because it simplifies deployment, as our school provides a web server where we can easily upload and host static HTML files. Additionally, this approach eliminates the need for server-side processing, reducing complexity and maintenance. The HTML report can also be viewed offline, making it more flexible.

4.4.6 File Handling & Path Management - Library Options

os Module

The os module is the standard choice for handling file paths, directory structures, and system commands in Python. It provides cross-platform compatibility, making it the most straightforward solution for managing file storage and retrieval. [47]

Pathlib

Pathlib is a modern alternative to os that introduces an object-oriented approach to file handling. While it offers improved readability, the os module remains more widely supported across different Python versions.[48]

Conclusion

The os module was chosen due to its broad compatibility and simplicity in handling file paths.

Implementation

5.1 Prototype

Lisa Kolli

5.1.1 Fundamental idea

Like already said in the Project definition [1.2.1] the hardware part of the project should be able to measure the electromagnetic radiation and store it. Additionally, it should also store the current location. To make this task, easier to execute, I divided the circuit in three parts.

The first part of the circuit has to amplify the signal, coming from the EMF antenna, since a ESP32 can only work with voltage from 0 V to 3.3 V.

After that the signal needs to be simplified, so that the device and we can work with it.

Last but not least there must be a circuit for the ESP32 and all the other components.

When everything works, the circuits will be combined.

5.1.2 Amplification circuit

Idea

There are a lot of ways to amplify a signal. One of the best-known technologies are operational amplifier (op-amp). I used the op-amp to build a not inverting amplifier. This kind of circuit provides a high input impedance, perfect for our application.

Operational amplifier

An operational amplifier is a DC-coupled amplifier with a high gain, which is completely dependent on the supply voltage range of the op-amp. Thanks to that no rectifier is needed. If the operational amplifier is only supplied with a voltage range from 0 V to 5 V, the minimum can be zero, which means that all the negative parts of the signal will be cut out and no longer be a problem for the ESP32.

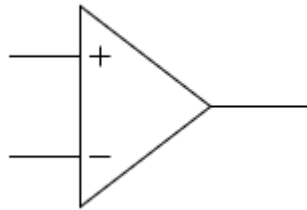


Figure 5.1:
operational
amplifier
schematic

For this circuit I used the LM358N/LM358P op-amp, which is to be exact a Low-Power, Dual-Operational Amplifier.

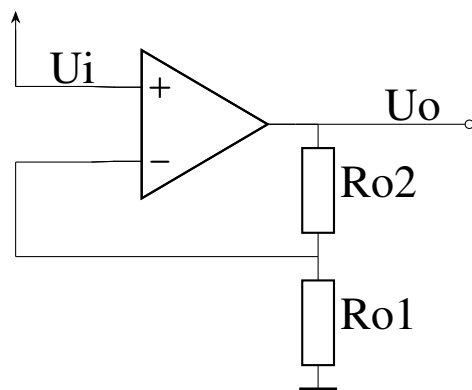
LM358N/LM358

In this chip there are two operational amplifier build in, however for this circuit only one is needed.

Supply voltage, V+	max. 32V
Minimum input voltage	-0.3V
Maximum input voltage	32V

[49]

Circuit



Components and values

op-amp	LM358N/LM358P
R _{o1}	1 kΩ
R _{o2}	22 kΩ
U _i	0 V - 120 mV
U _o	0 V - 5 V

Calculations

There were a lot of problems in the finding and calculating of the resistor values. The input signal always changes, dependent on the environment, circuit and unknown factors. Because of that there were a lot variations in the values of the resistors, almost every week a different one. As a result the following values equates the last and correct resistor values, which were found by measuring the input signal in the final product.

$$\frac{U_o}{U_i} = 1 + \frac{R_{o2}}{R_{o1}}$$

Forming Formulas to R_{o2}

$$R_{o2} = \frac{U_o * R_{o1}}{U_i} - R_{o1}$$

Inserting values $U_o = 3,3V$, $U_i = 150 \text{ mV}$, $R_{o1} = 1 \text{ k}\Omega$

$$R_{o2} = \frac{3,3 * 1 * 10^3}{0,15} - 1 * 10^3$$

$$R_{o2} = 21000\Omega = 21k\Omega$$

Chosen: $R_{o2} = 22 \text{ k}\Omega$

Gain-Bandwidth Product

Something I hadn't considered is the gain-bandwidth product. Ideally, an OPV would have no threshold frequency. However, in reality, every operational amplifier has a threshold frequency due to its inherent low-pass characteristics. This means that the higher the amplification, the lower the threshold frequency. It's also important to note that a higher threshold frequency usually comes with a higher price tag. In our case, the threshold frequency of the OPV is quite low.

The threshold frequency at an amplification of 23 is approximately 40 kHz.

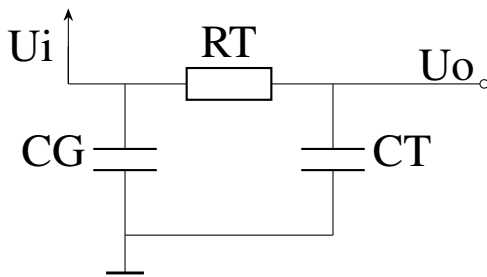
[49]

5.1.3 Intrinsic curve circuit

Idea

The next step is to create a circuit which makes the rather complicated and difficult to analyse signal, a lot more usable for us. This is needed because, there is no use in having a strongly fluctuating signal, especially with an ESP32. Therefore the task of this circuit is to form an envelope around the signal. [50]

Circuit



Components and values

CG	470 μ F
CT	220 pF
RT	120 m Ω
U _i	0 V - 5 V
U _o	0 V - 5 V

Smoothing capacitor

As a smoothing capacitor I used the capacitor CG. The function of the capacitor, is like the name already says, to smooth the curve of the input signal. The value of the capacitor was determined by testing.

Low-pass filter

RT and CT together form a low-pass filter, which cuts out all signals that are higher than the threshold frequency (f_g).

Calculations

$$fg = \frac{1}{2\pi * RT * CT}$$

Forming Formulas to CT

$$CT = \frac{1}{2\pi * RT * fg}$$

Inserting values $fg = 6 \text{ GHz}$ $RT = 120 \text{ m}\Omega$

$$CT = \frac{1}{2\pi * 120 * 10^{-3} * 6 * 10^9}$$

$$CT = 221 * 10^{-12} F = 221 pF$$

Chosen: $CT = 220 \text{ pF}$

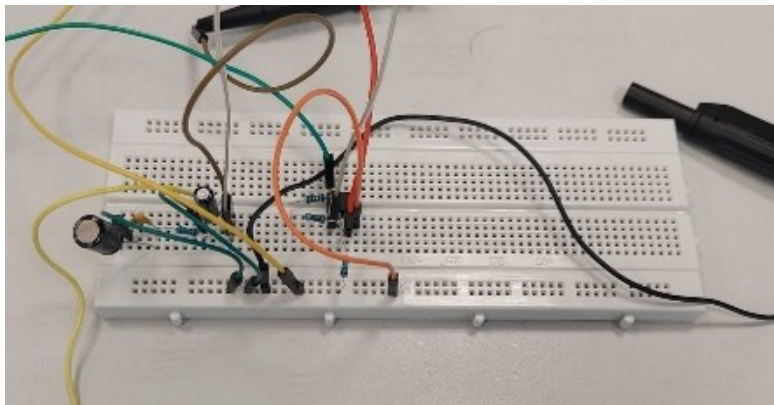
5.1.4 Combining

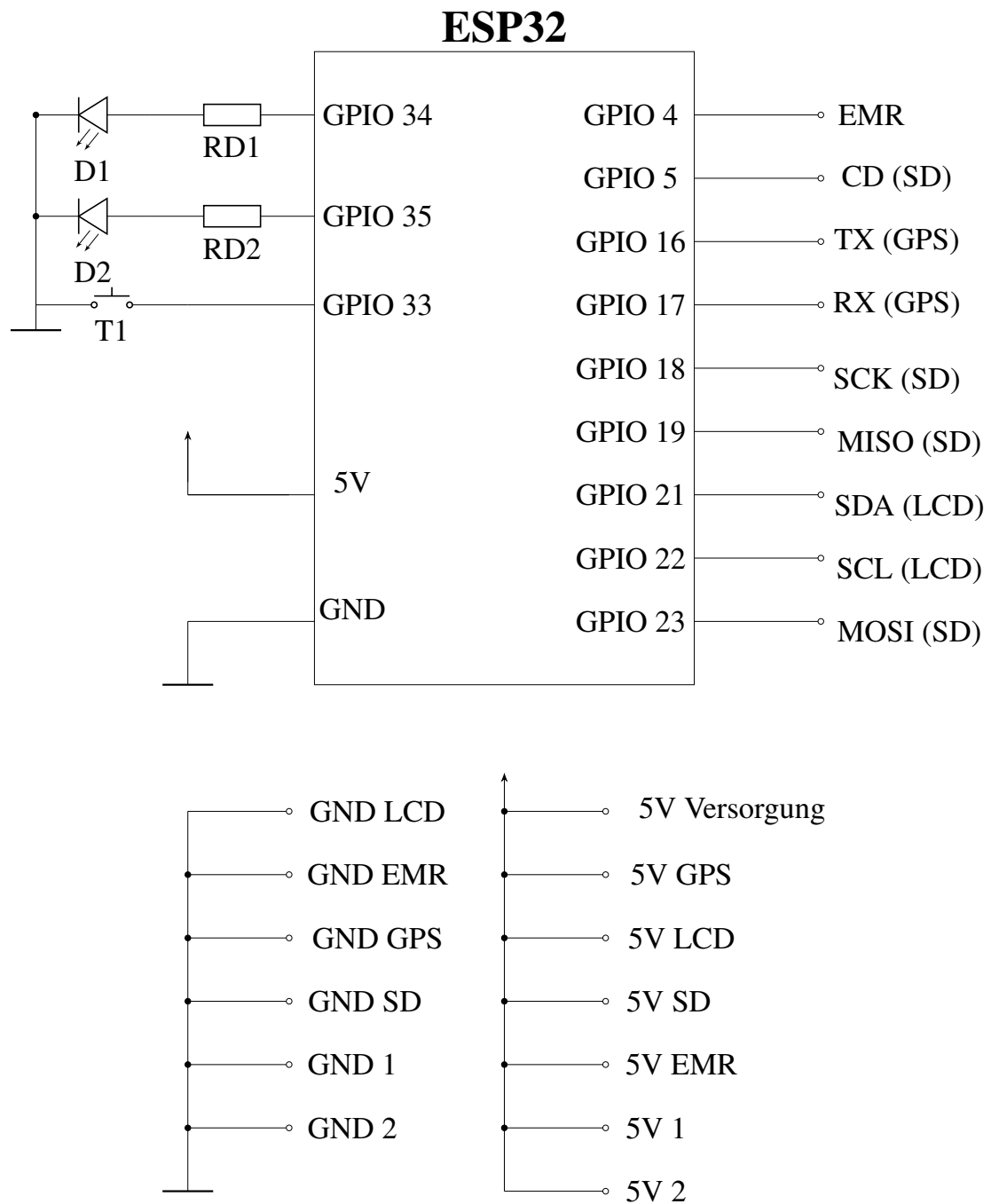
Figure 5.2:
First assembly on breadboard, amplify circuit and intrinsic curve circuit

The first test assembly was made on a breadboard.

5.1.5 ESP32 Circuit**Idea**

This circuit serves as a adaptation to the ESP32 microcontroller.

It is capable of receiving a GPS data and the data from the Circuit intrinsic curve, as well as show the data on a LC display. Additionally, there is also a button and two LEDs, for the use of the device.

Circuit

Components and values

D1	green
D2	red
EMR	
GPS	TX, RX, GND, 5V
LCD	SCL, SDA, GND, 5V
RD1	220 Ω
RD2	220 Ω
SD	CD, SCK, MISO, MOSI, GND, 5V

LEDs

D1 should show, if the device saves the electromagnetic radiation and the GPS data on the SD card. If it does, it should light up.

The light-emitting diode D2 should light up, when the device receives a GPS Signal. This should help the user of the device.

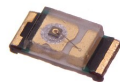


Figure 5.3:
Log periodic
antenna)

Button

T1 one is a push button with the purpose to activate or deactivate saving of the data onto the SD card.

SD card

The SD has a lot of pins in this circuit (CD, SCK, MISO, MOSI, GND, 5V). All those connections are needed for the SD module to work. Thanks to this module it is possible to save the important data (EMR, GPS) on a SD card.

GPS modul

Just like the SD module, the GPS module has a lot of connections to the circuit (TX, RX, GND, 5V). It allows us to locate the position of this device.

LCD

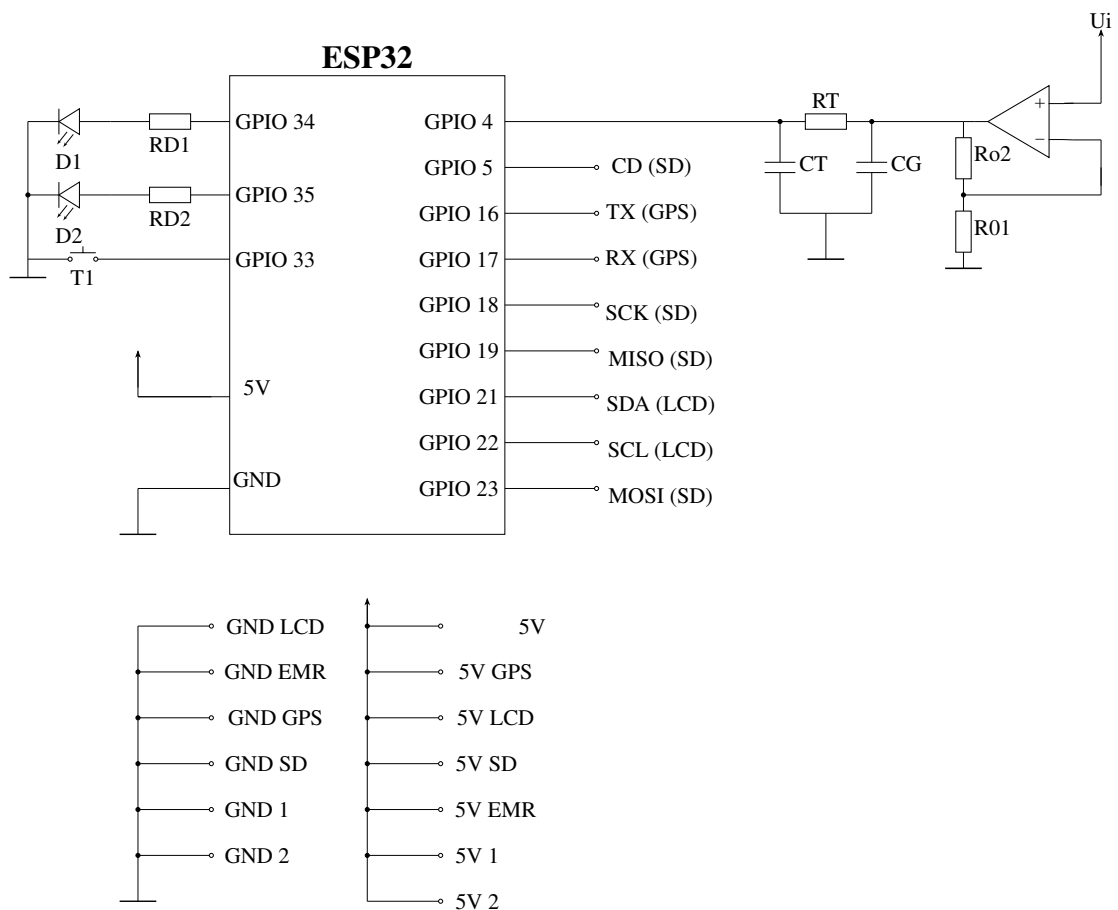
The Liquid Crystal Displays (LCD) is here to show the current EMR and GPS data. It has a lot connections to the circuit (SCL, SDA, GND, 5V).

EMR

There are many parts of the circuit called EMR. EMR stands for electromagnetic radiation and describes in this circuit the connections to the combination of the amplifying and intrinsic curve circuit. Through this connections the ESP32 should gain the size of the electromagnetic radiation.

5.1.6 Circuit combined

Circuit



Components and values

CG	470 μ F
CT	330 μ F
D1	green
D2	red
EMR	
GPS	TX, RX, GND, 5V
LCD	SCL, SDA, GND, 5V
op-amp	LM358N/LM358P
RD1	220 Ω
RD2	220 Ω
Ro1	1 k Ω
Ro2	22 k Ω
RT	120 Ω
SD	CD, SCK, MISO, MOSI, GND, 5V
T1	button
Ui	0 V - 120 mV

Function

This circuit does exactly what it is supposed to do. After uploading the Code [5.5] the display showed the EMF and the device was also able to save the EMF with the current location.

5.2.2 Component list

Lisa Kolli

CG	470 μ F
CT	330 μ F
D1	green
D2	red
EMR	
GPS	TX, RX, GND, 5V
LCD	SCL, SDA, GND, 5V
op-amp	LM358N/LM358P
Powerbank	
RD1	220 Ω
RD2	220 Ω
Ro1	Ω
Ro2	Ω
R3	10 M Ω
RT	120 Ω
SD	CD, SCK, MISO, MOSI, GND, 5V
T1	button
T2	switch
Ui	0 V - 120 mV

Switch

The switch (T2) has the function to connect the powerbank with the ESP32. The switch can also be called a power supply or on/of switch.

Pull down resistor (R3)

Since everything worked the circuit boards was created [5.3].

However, after PCB assembly, a problem occurred. Even though the antenna is connected and the software [5.5] works perfectly fine, the ESP32 always shows the maximum possible EMR. Due to the issue I decided to recreate the entire circuit on a patch panel and to my surprise it worked, even though I changed nothing. The only difference was, that I did not measure the input signal on the circuit board. After connecting a oscilloscope to the input of the circuit (where the antenna is) it worked.

After some research, I have come to the conclusion that the reason for the strange values, are that the antenna is too high-impedance. To resolve the issue I put an high resistant resistor (10 M Ω) between the antenna and ground, this kind of resistor is called a Pull down resistor.

5.3 Circuit design

Letissa Adlboller

5.3.1 Circuit-Board Overview

ESP-Circuit

The ESP-Circuit was designed to fit the ESP-Controller as well as several Pins in order to connect other components as well as the Amplifier-Circuit and the LED-Circuit to it. (more detailed information: 5.2.1)

Amplifier-Circuit

The Amplifier-Circuit was designed to amplify the received electromagnetic signal from the antenna using a non-inverting amplifier, as the signal on its own would be too weak and could cause problems for the ESP32. (more detailed information: 5.2.1)

LED-Circuit

The LED-Circuit was simply designed to light up LEDs indicating whether a GPS-Signal is received or not, as well as signaling if the current EMF-Data is saved on the SD-Card.

5.3.2 Design Process

Used Software

Altium Designer

All of the circuits were designed in the app Altium Designer. The app provides a user-friendly environment to create schematics and PCB-Designs as well as 3D-Modelling and much more. The component library includes a wide range of components which made it useful for our project. Altium provides all the tools that are needed to design a printable circuit board from start to finish.

JLCPCB

The circuit boards were ordered on the website from the global PCB manufacturer JLCPCB. The Manufacturer provides high-quality at a low cost and fast delivery, which made it fitting for our project. Moreover, JLC provides a final inspection of the manufactured boards which ensured the functionality of our circuits.

EMC PCB-Design Guidelines

Electromagnetic compatibility (EMC) is a characteristic of an electronic device, that allows it to operate as intended within an electronic environment, without generating Electromagnetic Interference (EMI) in nearby devices.

EMI is unwanted electromagnetic noise or interference which can cause electronics to operate poorly, malfunction or stop working completely. EMI is divided into two categories: conducted

emission (CE) and radiated emission (RE) [51]

Conducted emission (CE)

These emissions are transferred to another device through physical connections such as wires, PCB traces or power input lines. [52] Conducted emissions can occur in two forms, common-mode or differential mode both signals need two lines to distribute signals [53].

- Common-mode Signals:
Signals flow through the two lines in the same direction and Phase.
- Differential-mode signals:
Signals flow in opposite directions through two lines in close proximity.

Common sources of CE are:

- Power Supply
The Power Supply is a primary source of conducted EMI, as noise can travel throughout the PCB. Several factors such as voltage spikes, switching frequencies, and improper grounding techniques can amplify these issues.
- Micro-controllers
Microcontrollers are a source of conducted EMI due to their high-speed internal switching and their communication with other interfaces, which contain harmonics that can radiate EMI.
- Clock Signals
High-frequency clock signals can generate conducted and radiated EMI due to high-frequency switching.
- High-Speed Digital Signals
Digital Signals like those for communication contain harmonics that can transfer conducted EMI in the power lines.

Radiated emission (RE)

Radiated emissions travel through the air. They are caused when a device produces intentional or unintentional electromagnetic energy in the form of an electrical field. [54]

Common sources of RE are:

- Amplified noise
Radiated emission can come from external sources such as radio signals, telecommunication equipment, radar systems, electrical motors etc.

- Switching noise
Electronic that switch at a high frequencies including switch-mode power supplies, AC/DC Converters etc.
- Cavities
Cavities in the board due to poor stack up design or gaps in planes create electromagnetic resonance.
- poor PCB Routing
Sharp corner traces (90°) in a PCB disrupt the signal flow, causing the signal to reflect and create unwanted emission.

EMC (Electromagnetic compatibility) ensures that the circuit performs as intended and doesn't interfere with any other electronics in its surroundings. This ability is crucial for the project as any conducted or radiated EMI could distort the measured EMF value and could cause malfunctioning in the circuit.

Concerning the project, the following measures could have caused radiated or conducted EMI in the circuits:

- ESP (Microcontroller)
- High-Speed SPI Communications
- Power Supply
- poor Routing
- Clock Signal

Therefore, several steps were taken to avoid EMI in the PCB Layouts as far as possible [55][51][56].

Step 1. Grounding Planes

Signal traces which carry a current from one point to another must always have a proper returning path. Ground planes are optimal to ensure a direct low impedance return pathway for the current, back to the source. However, when routing over a split reference plane, the return current can not flow directly underneath the forward current causing unnecessary current loops. These current loops tend to lead to high radiated emission (RE). Which is why the Filter Circuit as well as the ESP-Circuit both have a solid ground plane on the top layer. Therefore, all the important SPI signals (MOSI, MISO, SCK, CS) are traced over this solid ground plane, ensuring proper returning paths for the signals and reducing noise!

Step 2. PCB trace routing

Signal routing is a key factor that needs to be considered when reducing emissions in a PCB-Design. High-speed signal traces must be carefully limited in their length, as longer lines can operate as antennas and emit electromagnetic interference. It is essential to avoid 90°degree angles in traces since they can lead to reflections and amplify EMI, as well as avoiding unnecessary loops. The project focused on keeping High-Speed Signals traces (MOSI, MISO, SCK, CS) short in length, using 45-degree angles for routing the traces as well as reducing loop area.

Step 3. Clock Signals

Clock Signals are due to their high frequency a common source of EMI, which is why its important to keep them as short as possible, which was also taken into account for this project.

Step 4. Microcontroller

Microcontrollers are known to cause Electromagnetic Interference (EMI) due to internal switching, Wi-fi and Bluetooth operations, Clock Signals, High-Speed Signals, improper grounding and so on. To reduce EMI from the ESP as far as possible proper grounding techniques such as a solid ground panel were used as well as making sure high-speed signal traces are kept short. In addition, we refrained from using Wi-Fi and Bluetooth communication with the ESP to minimize EMI even better.

Step 5. Shielding

To prevent EMI von entering or leaving the PCB, the whole circuit is shielded by an aluminum case. Therefore, electromagnetic noise can't interfere with the antenna and distort the measured EMF-Value. As the cable for the antenna is traced inside the case it is also shielded.

Schematics

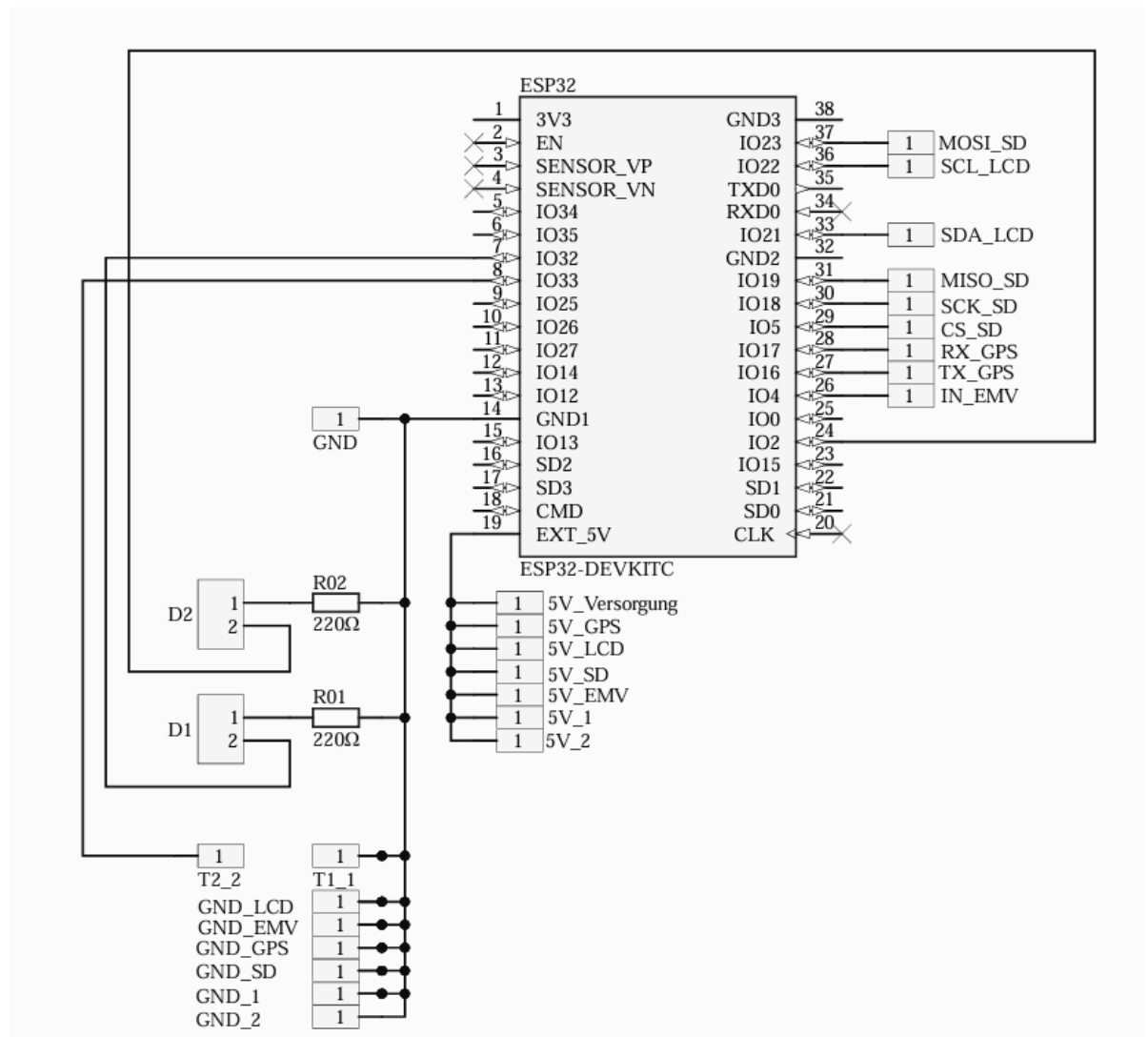


Figure 5.4:
Altium Schematic ESP-Circuit

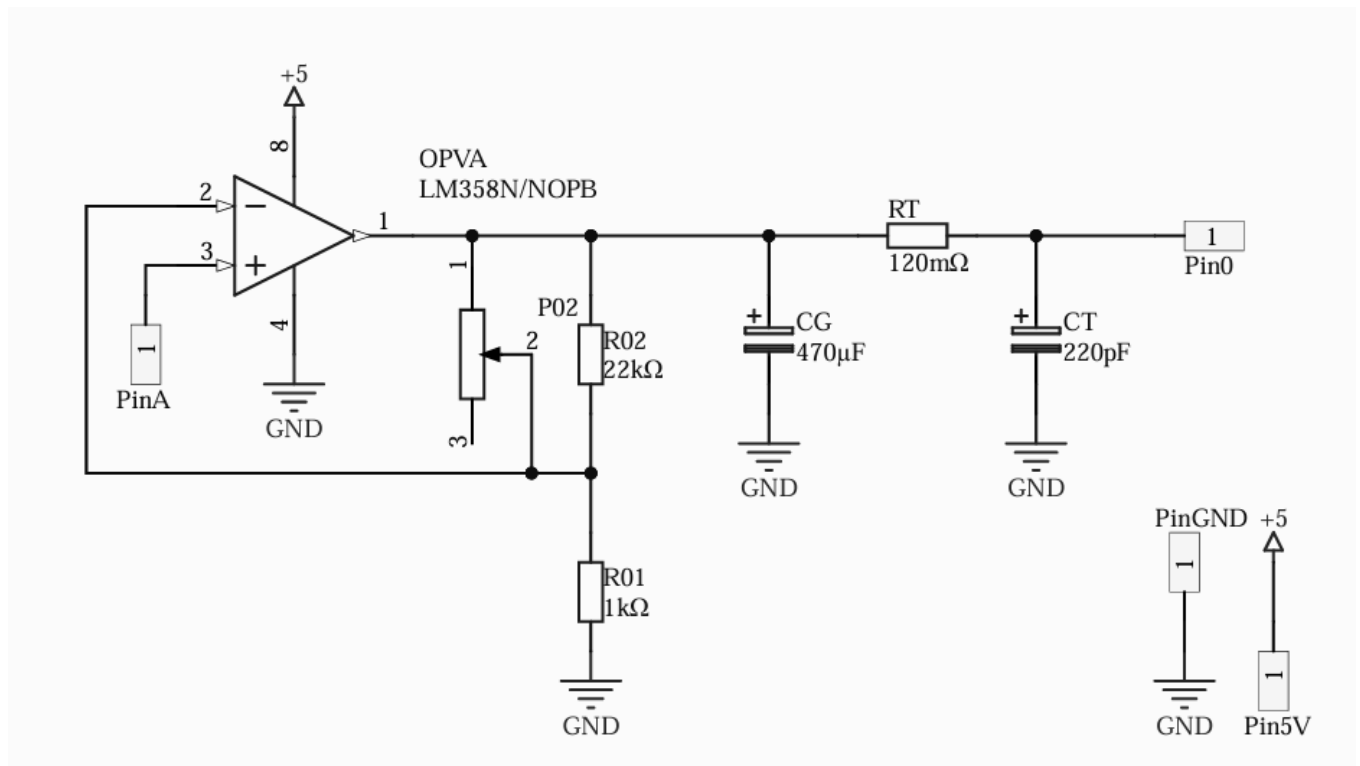


Figure 5.5:
Altium Schematic Filter-Circuit

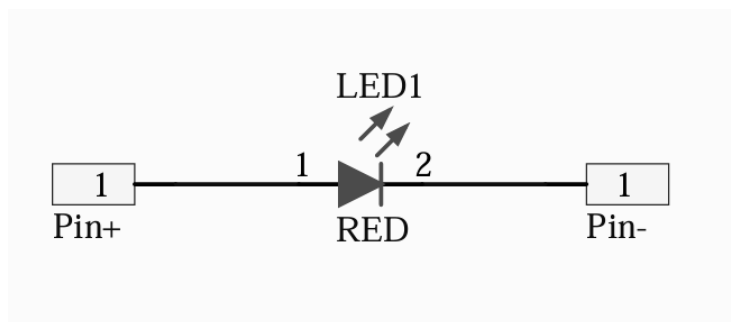


Figure 5.6:
Altium Schematic LED-Circuit

PCB-Layouts

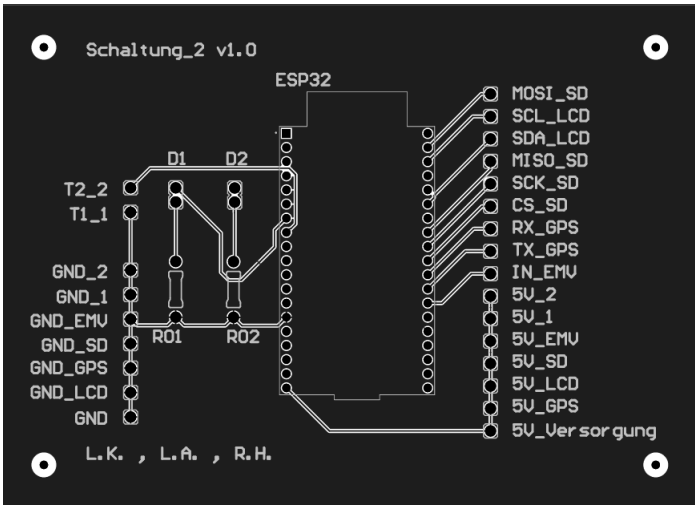


Figure 5.7:
Altium PCB-Layout
ESP-Circuit

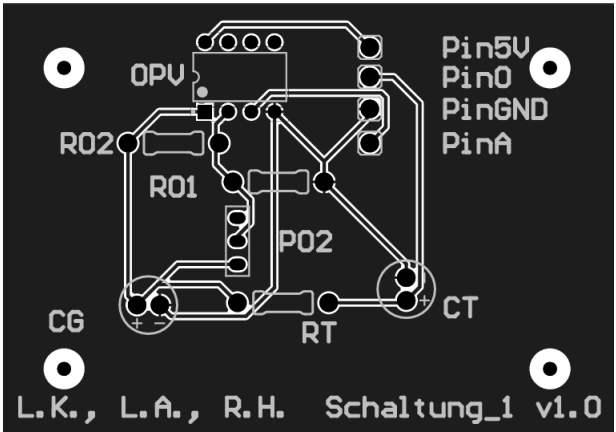


Figure 5.8:
Altium PCB-Layout
Filter-Circuit

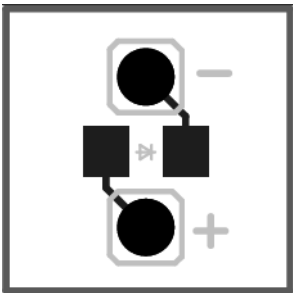


Figure 5.9:
Altium PCB-Layout
LED-Circuit

PCB 3D-Prints

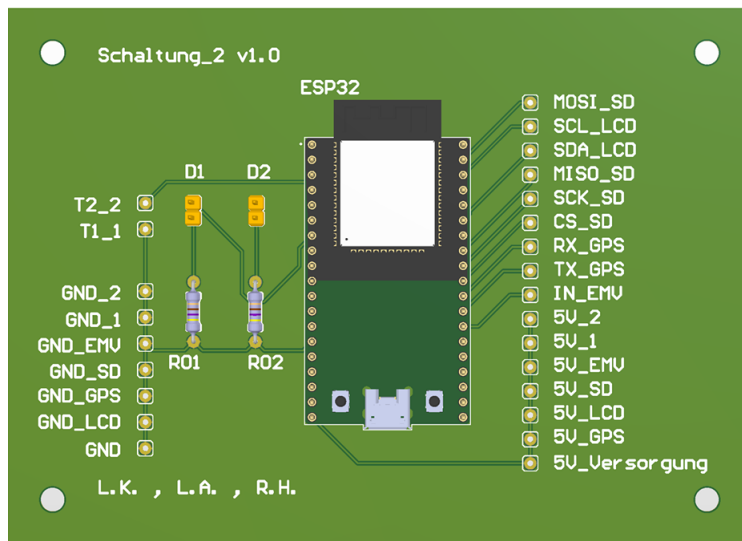


Figure 5.10:
Altium PCB 3D-Prints
ESP-Circuit

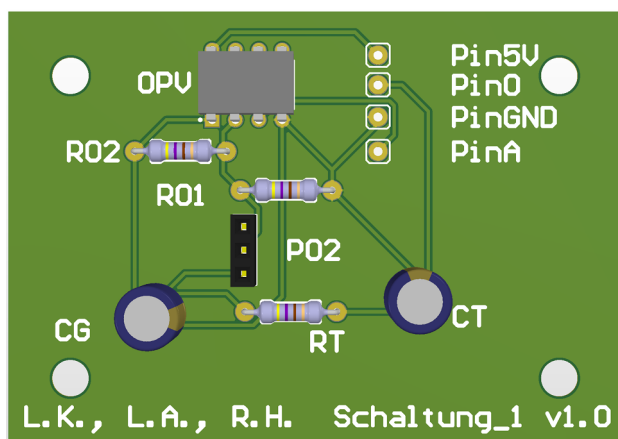


Figure 5.11:
Altium PCB 3D-Prints
Filter-Circuit

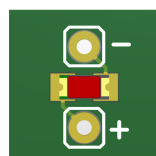


Figure 5.12:
Altium PCB 3D-Prints
LED-Circuit

Bill of Materials (BOM)

Comment	Description	Designator	Footprint	LibRef	Quantity
Pinheader 1x1	Pinheader 1x1	5V_1, 5V_2, 5V_EMV, 5V_GPS, 5V_LCD, 5V_SD, 5V_Versorgung, CS_SD, GND, GND_1, GND_2, GND_EMV, GND_GPS, GND_LCD, GND_SD, IN_EMV, MISO_SD, MOSI_SD, RX_GPS, SCK_SD, SCL_LCD, SDA_LCD, T1_1, T2_2, TX_GPS	PINHEADER_1X1	Pinheader 1x1	25
Pinheader, 2-Pin	Pinheader, 2-Pin	D1, D2	PINHEADER_1X2_2.54 mm_drill-1mm	Pinheader, 2-Pin	2
ESP32-DEVKITC	ESP32-WROOM-32UE series Transceiver; 802.11 b/g/n (Wi-Fi, WiFi, WLAN), Bluetooth ÅSmart Ready4.x Dual Mode Evaluation Board	ESP32	MODULE_ESP32- DEVKITC	ESP32-DEVKITC	1
Resistor	Ptot = 250mW	R01, R02	Resistor_Axial_1/4W_4 70R	Resistor_THT	2

Figure 5.13:
Altium BOM ESP-Circuit

Comment	Description	Designator	Footprint	LibRef	Quantity
470µF	Polarized Capacitor (ELKO, Aluminum ELKO, Tantal)	CG	CAPACITOR_POL_AL_R adial_D6.3	Capacitor Pol	1
220pF	Polarized Capacitor (ELKO, Aluminum ELKO, Tantal)	CT	CAPACITOR_POL_AL_R adial_D6.3	Capacitor Pol	1
LM358N/NOPB	Low Power Dual Operational Amplifier, 8-pin MDIP, Pb-Free	OPV	P0008A	CMP-0055-00153-3	1
Potentiometer_DUMMY	Potentiometer_DUMMY	P02	Potentiometer	Potentiometer_DUMMY	1
Pinheader 1x1	Pinheader 1x1	Pin0, Pin5V, PinA, PinGND	PINHEADER_1X1	Pinheader 1x1	4
Resistor	Ptot = 250mW	R01, R02, RT	Resistor_Axial_1/4W_4 70R	Resistor_THT	3

Figure 5.14:
Altium BOM Filter-Circuit

Comment	Description	Designator	Footprint	LibRef	Quantity
RED	LED SMD 1206 RED		LED_1206_RED	LED_SMD_1206_RED	1
Pinheader 1x1	Pinheader 1x1	~, +	PINHEADER_1X1	Pinheader 1x1	2

Figure 5.15:
Altium BOM LED-Circuit

Generating Manufacture Outputs

Generating output files

To prepare the manufacturing data of the circuit boards for the website JLCPCB the following files from the Altium project were needed:

- Gerber File
- NC Drill File

The best way to manage and generate these files was through an output job file [57]. An output job file is a pre-configured set of outputs which must be included into the project. It then allows one to configure each output with its own settings and its own format. There are several outputs to pick from, for this project the following outputs were used:

- Documentation Outputs:
were used to generate prints of the Schematic, PCB-Design, 3D PCB-Design
- Fabrication Outputs:
were used to generate the NC Drill Files and Gerber Files
- Report Outputs:
were used to generate the Bill of Material (BOM)
- Validation Outputs:
were used to generate the Design Rule Check

The outputs can then be written into three different types of output containers – a PDF (For Schematic Prints, PCB Prints, ...), a Folder Structure(for Gerber Files, NC Drill Files...) or a video. The container is then created at the chosen location either at a local path or a path in the server.

5.3.3 Challenges and Solutions

Missing Bottom Layer

The first PCB-Layouts were ordered incorrectly, instead of ordering Two-Layer boards with a Top- and Bottom-Layer, we accidentally ordered Single-Layer boards with only a Top-Layer.

Only after receiving the finished product did we notice that the Bottom-Layer on both boards was missing completely. There were two options to choose from to handle this Problem both with up- and downsides

Option 1: Not ordering new circuit boards

- complicated assembling of the board (soldering only on Top-Layer)

- incomplete appearance of the circuit board
- saving money by not ordering new boards
- continuing project as normal

Option 2: Ordering new circuit boards

- spending money on new boards
- waiting for the arrival of the product
- easy assembling of the boards (soldering as normal on Bottom-Layer)
- finished and clean look of the boards

Solution

Even though option 1 would have been easier than option 2, we picked option 2. It was crucial for the project to work properly by soldering all the components on the Top-Layer, this wasn't assured. Moreover, the overall appearance of the project was also an important factor for the group, a missing bottom-layer would have looked incomplete.

Another factor that contributed to this decision was the fact that we were ahead of our planned schedules, which is why waiting for the arrival wasn't a critical factor that could have majorly set the project back.

Also, the effort to order new boards wasn't big, as the problem was the wrong order not an incomplete layout.

5.3.4 Verification and Testing

Self testing

The Design Rule Check (DRC) is an automated feature created by Altium Designer, which allows users to test PCB-Layouts for both the logical and physical integrity of a design. It creates a Design Rule Verification Report which confirms whether minimum clearance rules have been maintained and checks for design violations[58].

This Design Rule Check was also used for the PCB-Layouts (ESP-Circuit, Filter-Circuit) of this project to assure the functionality of the circuits before manufacturing them.

Some of these Rules were altered due to the following reasons:

*Applies for both circuits

Hole size constraint

original: min= 0.025mm max= 2.54mm altered: min= 0.025mm max= 4mm Reason: The size of the four mounting holes on the PCB-Layout were marked as violations because their size is 4mm which is bigger than the original maximum value. Since these holes are only used for mechanical mounting and no other hole size was violated other than these four, the maximum size was altered to 4mm.

Silk to Solder Mask (checking for clearance to exposed copper)

original: min= 0.254mm altered: min= 0.15mm Reason: The pinheads on the PCB-Layout have a silkscreen frame around them, which violates the minimum silk to pad (exposed copper) clearance value. Since the manufacturer JLCPCB has the capability to produce with a minimum distance of 0.15mm between a pad and silkscreen, the value was altered to 0.15mm [59].

*Applies only for Filter-Circuit

Width Constrain

original: min= 0.254mm max= 0.254mm preferred= 0.254mm altered: min= 0.254mm max= 0.381mm preferred= 0.254mm Reason: Due to a tiny mistake, the traces in this PCB-Layout were accidentally routed with a wrong width (0.381mm). However, as there is no constriction for a maximum width for traces by the manufacturer JLPCB the value was altered to 0.381mm.

The LED-Circuit was not checked by the Design Rule Check due to the simplicity of the layout this wasn't necessary also this circuit wasn't produced by a manufacturer but with machines provided by the school.

Design Rule Verification Report -ESP-Circuit**Design Rules Verification Report**

Filename : C:\Users\letis\OneDrive - Höhere technische Bundeslehranstalt Graz-Göting\D

Warnings 0

Rule Violations 0

Warnings	
Total	0

Rule Violations	
Clearance Constraint (Gap=0.254mm) (All),(All)	0
Short-Circuit Constraint (Allowed=No) (All),(All)	0
Un-Routed Net Constraint (All)	0
Modified Polygon (Allow modified: No), (Allow shelved: No)	0
Width Constraint (Min=0.254mm) (Max=0.254mm) (Preferred=0.254mm) (All)	0
Routing Topology Rule(Topology=Shortest) (All)	0
Power Plane Connect Rule(Relief Connect)(Expansion=0.508mm) (Conductor Width=0.254mm) (Air Gap=0.254mm)	0
Hole Size Constraint (Min=0.025mm) (Max=4mm) (All)	0
Hole To Hole Clearance (Gap=0.254mm) (All),(All)	0
Minimum Solder Mask Sliver (Gap=0.254mm) (All),(All)	0
Silk To Solder Mask (Clearance=0.15mm) (IsPad),(All)	0
Silk to Silk (Clearance=0.254mm) (All),(All)	0
Net Antennae (Tolerance=0mm) (All)	0
Height Constraint (Min=0mm) (Max=25.4mm) (Preferred=12.7mm) (All)	0
Total	0

Design Rule Verification Report -Filter-Circuit

Design Rules Verification Report

Filename : C:\Users\letis\Downloads\Filterschaltung\Filterschaltung\PCB_Filter.PcbDoc

Warnings 0
Rule Violations 0

Warnings	
Total	0
Rule Violations	
Clearance Constraint (Gap=0.254mm) (All),(All)	0
Short-Circuit Constraint (Allowed=No) (All),(All)	0
Un-Routed Net Constraint (All)	0
Modified Polygon (Allow modified: No), (Allow shelved: No)	0
Width Constraint (Min=0.254mm) (Max=0.381mm) (Preferred=0.254mm) (All)	0
Routing Topology Rule(Topology=Shortest) (All)	0
Power Plane Connect Rule(Relief Connect)(Expansion=0.508mm) (Conductor Width=0.254mm) (Air Gap=0.254mm)	0
Hole Size Constraint (Min=0.025mm) (Max=4mm) (All)	0
Hole To Hole Clearance (Gap=0.254mm) (All),(All)	0
Minimum Solder Mask Sliver (Gap=0.254mm) (All),(All)	0
Silk To Solder Mask (Clearance=0.15mm) (IsPad),(All)	0
Silk to Silk (Clearance=0.254mm) (All),(All)	0
Net Antennae (Tolerance=0mm) (All)	0
Height Constraint (Min=0mm) (Max=25.4mm) (Preferred=12.7mm) (All)	0
Total	0

Quality Tests by Manufacturer JLCPCB

Quality Control

JLCPCB offers several quality controls as part of their production process in every aspect of the manufacturing.

- Firstly, the submitted Gerber Files are being reviewed by engineers, confirming whether the design remains true on tolerance restrictions, spacing between pads, traces etc. before sending them to production.
- They ensure maximum quality control over materials as well as advanced equipment.
- Before shipping multiple tests are carried out: First AOI (Automated Optical Inspection) second, Flying Probe Testing and lastly final Quality Checking piece by piece before shipping.

After self-testing the PCB-Layouts and after several quality control checks by the manufacturer the Circuit-boards are fully functional and can be used for the project.

5.4 Case

Lisa Kolli

To make the device look more appealing and to protect it from any kind of environmental factors, a case must be made.

The case should keep the EMR outside and also make sure, that the EMR from our hardware does not interfere with the received signal from the log periodic antenna. For that reason a main part of the housing is made out of aluminium.

Additionally the case is also made with 3D printed parts, made out of PLA.

The device must be portable and arrangeable. It would also be practical, if the antennas are removable, since that makes the storing and transporting easier.

5.4.1 Software

AutoCAD

To design the 3D printed parts, I decided to use AutoCAD, a 2D and 3D CAD software application developed by Autodesk. [60] AutoCAD is also a Vector drawing program and probably one of the best known CAD programs used in architecture, engineering, interior designing, graphic designing, city planning, and so much more.

Ultimaker Cura 5.9.0

After 3D modelling the parts in AutoCAD, they also need to be printed. For that the printer Anycubic Kobra, however before printing the stl file needs to be converted to a gcode file. To do this we used Ultimaker Cura 5.9.0, an open source slicer software. [61]

5.4.2 Placement

On the front the Display, two LEDs and one button are located. Positioned on the top of the case is the antenna and on the bottom are the main power switch, SD Card slot and the charging socket. Last but not least is the GPS antenna, which is located on the right side of the housing.

5.4.3 3D printing

For the 3D printing, I used two different filaments, one was black, and the other one was transparent. Overall, the case consists of 13 3D printed parts.

Back filament

Altogether 11 parts were printed with black filament.

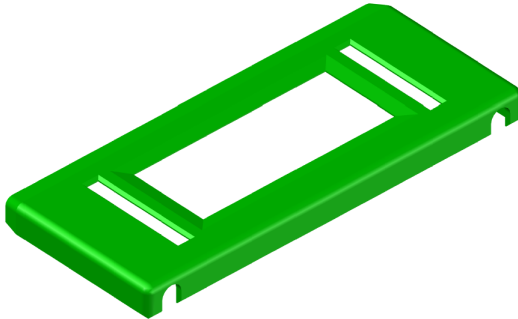


Figure 5.16:
3D part on the front of the
device

All in this section and flowing sections not used pictures are in the appendix. [B.1]

Transparent filament

The transparent filament was used to print two parts. These two 3D prints were used as a diffusor for the two used LEDs.

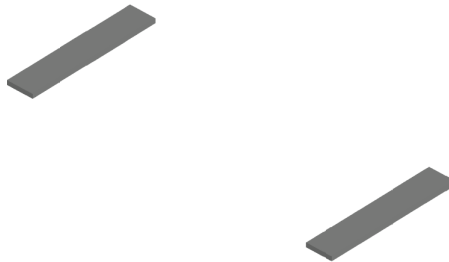


Figure 5.17:
LED diffusers

5.4.4 Antenna mount

The Antenna mount was the part of the housing, that caused the most challenges. The problem with the antenna is, that the antenna is quite heavy and the rod of the antenna could break, if not supported.

First design

Since the Antenna should be removable, we designed the mount so, that the antenna gets screwed directly into the housing. However because of that removing and adding the antenna was quite difficult. In addition the screw-on part of the antenna is the most sensible part of the antenna and it's cable. It didn't took long for the screw-on part of the cable to break.

Second design

After the cable broke I decided to take a different approach. Instead of building the screw-on part in the aluminium housing, I moved the screw-on part of the antenna outside. because of that the antenna mount needs to be longer, which makes it automatic a little bit bigger. Additionally the 3D printed part on the top of the device also needed to be changed, since the antenna mount changed. To attach the antenna to the mount, a metal nut got glued into the mount and knurled screws are used.

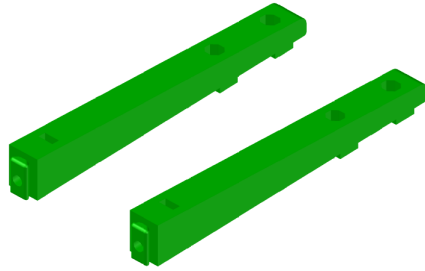


Figure 5.18:
Antenna mount

5.4.5 GPS antenna mount

The GPS antenna was the last bought component, it arrived after most of the housing already was finished. Because of that it didn't quite fit on the case of that the device, the solution was a false edge for the antenna. Given that the antenna is also magnetic, I glued a metal flat washer into the 3D printed false edge.

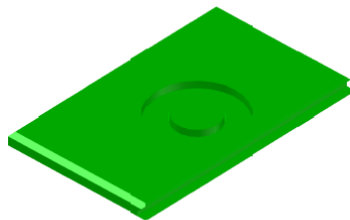


Figure 5.19:
GPS
Antenna mount

5.4.6 USB-C socket

After testing, the USB-C socket did not work and I had problems with the cable duct. As a result a new USB-C socket was bought, however it is a lot bigger than the first one, this required a 3D print. The bottom part of the device needed a bigger hole for the USB-C socket, as a consequence the SD card slot had to go further down. After implementing the changes, everything worked.

5.4.7 Assembling

Before assembling the case, I needed to cut the for the components needed holes into the aluminium housing. To mark where the hole and how big the hole is, I used the 3D printed parts. I laid them onto the aluminium housing and used them as a gauge. After cutting the holes the parts got glued together.

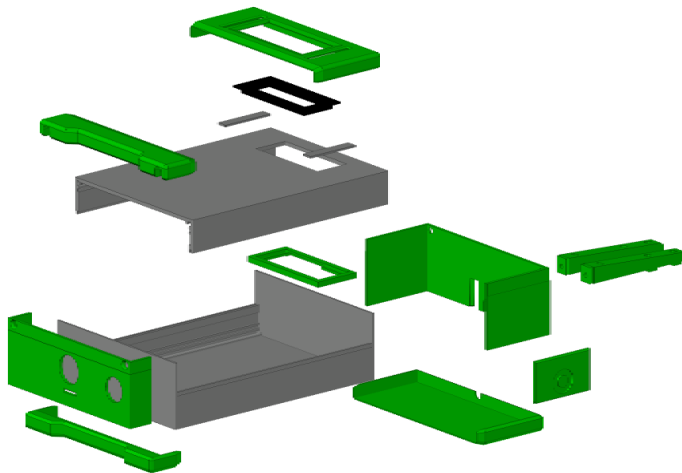


Figure 5.20:
Case with all parts

5.4.8 Adhesive

To glue together different types of adhesive, were being used. Most of the connections between the 3D printable parts and the aluminium housing, were agglutinated with double-faced adhesive tape. If something come of and using the double-faced adhesive tape to put it back on did not work, I used silicone to glue it to the aluminium. If 3D parts needed to be connected with each other, I used a special adhesive for PLA.

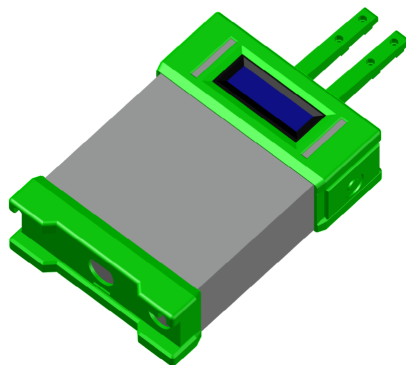


Figure 5.21:
Finished Case

5.5 Software

Robin Heindl

In this section, the focus is on transforming the project concept into a functional software solution. The software implementation covers two main components: the firmware running on the ESP32 (implemented in C/C++) and the data processing/visualization module developed in Python. While the hardware structure was completed by Lisa Kolli, this section details how the software bridges the gap between sensor data acquisition and the final user-facing interactive dashboard.

5.5.1 Initial Testing Phase

Before integrating real GPS coordinates and electromagnetic radiation (EMR) values, I established a controlled testing environment using simulated data. In this phase, EMR values were generated randomly within a range of 0 to 144 mV to represent realistic variations in electromagnetic fields. To maintain geographical plausibility, the simulated GPS coordinates were constrained to the Graz region, ensuring that the test data did not contain any unrealistic or extreme values. The generated test values were stored in a CSV file, which allowed for thorough testing of the Python visualization independently of the hardware setup. The initial version of the Python program was deliberately kept minimal. Its sole purpose was to generate a heatmap from the CSV data. This early prototype helped evaluate the visual representation of the data and ensured that the file handling and mapping functionalities operated as expected. Throughout this phase, various visualization techniques were explored and the heatmap rendering was optimized, drawing inspiration from resources like the Real Python Folium Guide [62].

5.5.2 ESP32 Firmware Software

The firmware for the ESP32 was developed in C/C++ and makes extensive use of libraries to manage its diverse responsibilities. The code is structured modularly, with each section dedicated to a specific task. Below, key sections are described in detail, including explanations of relevant code snippets.

Initialization

The firmware begins by including necessary libraries such as `LiquidCrystal_I2C`, `SdFat`, `SPI`, and `TinyGPS++`. Along with these inclusions, hardware-specific configurations are defined. The pins for the LED indicators, the button, and the GPS serial communication are established at the very start of the code. This early configuration ensures that all hardware components are correctly initialized before any further processing occurs.

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <SdFat.h>
#include <SPI.h>
#include <TinyGPS++.h>

// Defines RX and TX pins for Serial 2 + GPS baud rate
#define RXD2 16
#define TXD2 17
#define GPS_BAUD 9600

// Pin configuration
const int chipSelect = 5;           // SD card chip-select pin
const int emfPin = 4;              // Analog pin for the EMF sensor
const int greenLEDPin = 32;        // Green LED for "Loading"
const int redLEDPin = 2;           // Red LED for "GPS signal received"
const int buttonPin = 33;          // Button to start/stop saving
bool isLoading = false;            // Variable to store loading state
unsigned long outputTime = 0;       // To delay LCD and serial output
unsigned long outputDelay = 2000;   // Output delay time
unsigned long gpsSignalTime = 0;    // Time of last GPS signal
unsigned long lastSaveTime = 0;     // Time of last save
unsigned long saveInterval = 1000; // Interval of 1 second (1000 ms)
const unsigned long gpsLedDuration = 500; // LED blink duration in ms
bool previousButtonState = false;   // Tracks previous button state
```

Figure 5.22:
Initialization Code

In the `setup()` function, the first steps involve initializing the LCD through the `setupLCD()` function, which not only turns on the backlight but also displays a startup message to inform the user that the device is beginning its operation. Next, the SD card is initialized by calling `setupSDCard()`. Successful initialization of the SD card is crucial because it allows the program to create and write a CSV file. The CSV header is subsequently written using `writeCSVHeader()`. Finally, the secondary serial port is configured to handle incoming GPS data via the `TinyGPS++` library.

```
void setup() {  
  // Serial communication for debugging  
  Serial.begin(115200);  
  
  // Initialize LCD and SD card  
  setupLCD();  
  setupSDCard();  
  
  // Write CSV header  
  writeCSVHeader();  
  
  // Configure LED pins as output  
  pinMode(greenLEDPin, OUTPUT);  
  pinMode(redLEDPin, OUTPUT);  
  
  // Configure button as input with pull-up resistor  
  pinMode(buttonPin, INPUT_PULLUP);  
  
  // Initialize loading state  
  digitalWrite(redLEDPin, LOW); // Start with "Not loading"  
  digitalWrite(greenLEDPin, LOW);  
  
  // Start Serial 2 with defined RX & TX pins and baud rate  
  gpsSerial.begin(GPS_BAUD, SERIAL_8N1, RXD2, TXD2);  
  Serial.println("Serial 2 started at 9600 baud rate");  
  
  delay(2000); // Wait to show display  
  lcd.clear();  
}
```

Figure 5.23:
Setup()
Function

Main Loop and Data Logging

After initialization, the firmware enters its main loop, which updates sensor readings, displays current values, and logs data when appropriate.

```
void loop() {  
  // Check button state and toggle loading  
  checkButton();  
  
  // Retrieve GPS values  
  GPS();  
  
  // Read EMF value from the sensor on analog pin 4  
  emfValue = readEMF();  
  
  int Millis = millis();  
  if ((Millis - outputTime) > outputDelay) {  
    // Display values on LCD and serial monitor  
    displayValues(xValue, yValue, emfValue);  
  
    Serial.print("X: ");  
    Serial.print(xValue, 2);  
    Serial.print(", Y: ");  
    Serial.print(yValue, 2);  
    Serial.print(", EMF: ");  
    Serial.print(emfValue); // EMF value without decimals  
    Serial.println(" mV");  
  
    outputTime = millis();  
  }  
}
```

Figure 5.24:
Beginning
of Loop

At the beginning of each iteration, the `checkButton()` function is called. This function reads the state of a button (using a debounce mechanism) to determine whether data logging should be active. When the button is pressed and then released, it toggles the loading mode, enabling or disabling data collection.

```
// Check button state with debounce
void checkButton() {
    bool currentState = !digitalRead(buttonPin);
    // Read button (inverted due to INPUT_PULLUP)

    if (!currentState && previousButtonState) {
        // Change state only when button is released
        isLoading = !isLoading;
    }

    previousButtonState = currentState;
}
```

Figure 5.25:
CheckButton()
Function

Simultaneously, the GPS() function runs to update the current GPS coordinates. This function checks whether new and valid GPS data is available and updates global variables accordingly (for example, if `gps.location.isUpdated()` returns true, the latest coordinates are stored in `xValue` and `yValue`). Meanwhile, the `readEMF()` function reads the analog input from the EMF sensor, converts this raw data into millivolts, and stores the result in `emfValue`. I gathered my information to this topic through a GPS-Module knowhow page [63] and through reading the Library notes.

```
void GPS() {  
  // Read GPS data  
  while (gpsSerial.available() > 0) {  
    gps.encode(gpsSerial.read());  
    if (gps.location.isValid()) {  
      gpsSignalTime = millis();  
    }  
  }  
  
  if (gps.location.age() > 2000) {  
    digitalWrite(redLEDPin, LOW); // LED off  
  } else {  
    unsigned long currentMillis = millis();  
    if ((currentMillis - gpsSignalTime) % (2 * gpsLedDuration) < gpsLedDuration) {  
      digitalWrite(redLEDPin, HIGH); // LED on  
    } else {  
      digitalWrite(redLEDPin, LOW); // LED off  
    }  
  }  
  
  if (gps.location.isUpdated()) {  
    xValue = gps.location.lng();  
    yValue = gps.location.lat();  
  }  
}
```

Figure 5.26:
GPS()
Function

Every few seconds, the current GPS coordinates and EMR values are displayed on the LCD. If the device is in logging mode and valid GPS data is present, the program writes the data to a CSV file using `saveToCSV()`. A green LED lights up during this process.

```

if (isLoading) {
    // Green LED lights up while loading
    digitalWrite(greenLEDPin, HIGH);
    if (gpsSerial.available() > 0) {
        gps.encode(gpsSerial.read());
        // Check if GPS data is valid and enough time has passed since last save
        if (isValidGPS() && (millis() - lastSaveTime >= saveInterval)) {
            if (xValue > 0 && yValue > 0) {
                // Save values to CSV file
                saveToCSV(xValue, yValue, emfValue);

                // Update last save time
                lastSaveTime = millis();
            }
        }
    }
} else {
    // Green LED turns off when not loading
    digitalWrite(greenLEDPin, LOW);
}
}

```

Figure 5.27:
Loading,
part
of Loop

	A
1	
2	X,Y,EMF
3	15.400925,47.103386,20
4	15.400949,47.103477,20
5	15.400969,47.103508,20
6	15.400978,47.103542,20
7	15.400999,47.103561,20
8	15.401036,47.103630,20
9	15.401054,47.103638,20
10	15.401072,47.103657,20

Figure 5.28:
data.csv file

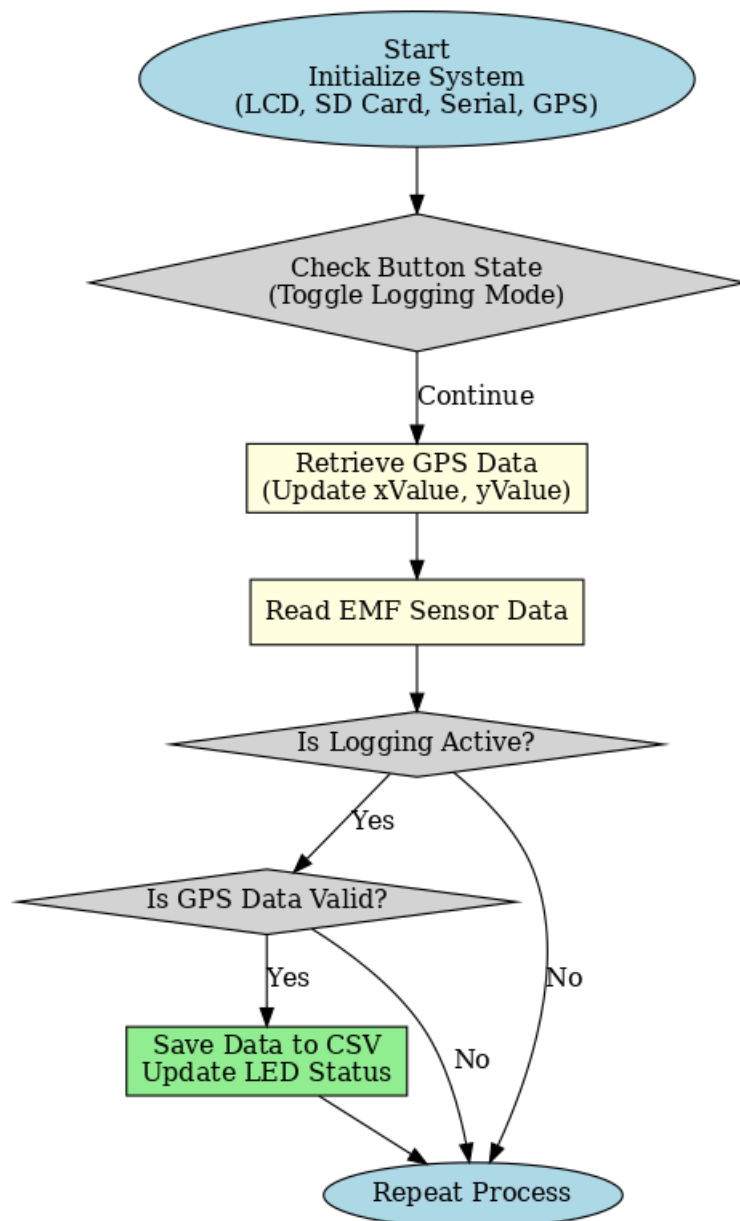
Firmware Flowchart

Figure 5.29:
Firmware Flowchart

The firmware operation begins with initializing essential components, including serial communication, the LCD display, the SD card, and the CSV file header. Once initialized, the system continuously executes key functions.

First, the `checkButton()` function monitors the button state, allowing the user to toggle logging mode. The firmware then retrieves GPS coordinates via `GPS()` and reads EMF values using `readEMF()`, displaying the data on the LCD with `displayValues()`.

If logging is enabled and valid GPS data is available (`isValidGPS()`), the values are written to a CSV file using `saveToCSV()`, and a green LED indicates successful recording. If logging is off, the LED remains off. This cycle repeats continuously, ensuring real-time data collection.

5.5.3 Python Data Processing and Visualization

Once the CSV file is generated by the ESP32, the Python module reads, processes, and visualizes the data, generating an interactive HTML dashboard.

CSV Data Import and Cleaning

The Python script starts by importing the necessary modules and setting up file paths using the `os` module. The CSV file, which contains the collected EMR and GPS data, is then read into a Pandas DataFrame. To ensure the data is usable, the script converts the GPS coordinates and EMR values to numeric types and drops any rows that contain invalid or missing data. This cleaning process is crucial for creating a robust dataset that is ready for subsequent analysis.

```
# Define paths based on current directory
current_directory = os.path.dirname(os.path.abspath(__file__))
csv_path = os.path.join(current_directory, "data.csv")
html_output_path = os.path.join(current_directory, "index.html")

# Check if CSV file exists
if not os.path.exists(csv_path):
    raise FileNotFoundError(f"The file 'data.csv' was not found in {current_directory}.")

# Load CSV data into a DataFrame and clean the data
df = pd.read_csv(csv_path, header=0)
df.columns = ['Longitude', 'Latitude', 'EMR']
df['Latitude'] = pd.to_numeric(df['Latitude'], errors='coerce')
df['Longitude'] = pd.to_numeric(df['Longitude'], errors='coerce')
df['EMR'] = pd.to_numeric(df['EMR'], errors='coerce')
df = df.dropna(subset=['Latitude', 'Longitude', 'EMR'])
```

Figure 5.30:
CSV-Data
Import
and
Cleaning

Visualization Generation

With the data cleaned, the next step is to visualize it. The script uses the Folium library to create an interactive heatmap. To make the EMR values visually intuitive, they are first scaled relative to the maximum observed value. The heatmap uses a color gradient (from green to red) to indicate the intensity of the electromagnetic radiation.

```
# Scale EMR values for visualization
df['EMR_Scaled'] = df['EMR'] / df['EMR'].max()

# Create a map centered around the average GPS coordinates
mymap = folium.Map(location=[df['Latitude'].mean(), df['Longitude'].
mean()], zoom_start=12)

# Prepare heatmap data as a list of [latitude, longitude, scaled EM
R]

heat_data = [[row['Latitude'], row['Longitude'], row['EMR_Scaled']]
for index, row in df.iterrows()]

# Generate the heatmap and add it to the map
HeatMap(heat_data, radius=25, blur=15, max_zoom=1).add_to(mymap)

# Add a linear color map (legend) to the map
colormap = LinearColormap(colors=['green', 'yellow', 'red'], vmin=0
, vmax=144)
colormap.add_to(mymap)

# Store the HTML representation of the map
map_html = mymap._repr_html_()
```

Figure 5.31:
Visualization
Code

In addition to the heatmap, the script generates three different types of charts using Matplotlib: a bar chart, a line chart, and a boxplot. Each chart is converted into a Base64-encoded string, enabling its direct embedding in the HTML file. For instance, the conversion function is defined as follows:

```
def fig_to_base64(fig):
    buf = io.BytesIO()
    fig.savefig(buf, format='png', dpi=300, bbox_inches='tight')
    buf.seek(0)
    encoded = base64.b64encode(buf.read()).decode('utf-8')
    buf.close()
    return f"data:image/png;base64,{encoded}"
```

Figure 5.32:
Base64 conversion
function

Python Module Flowchart

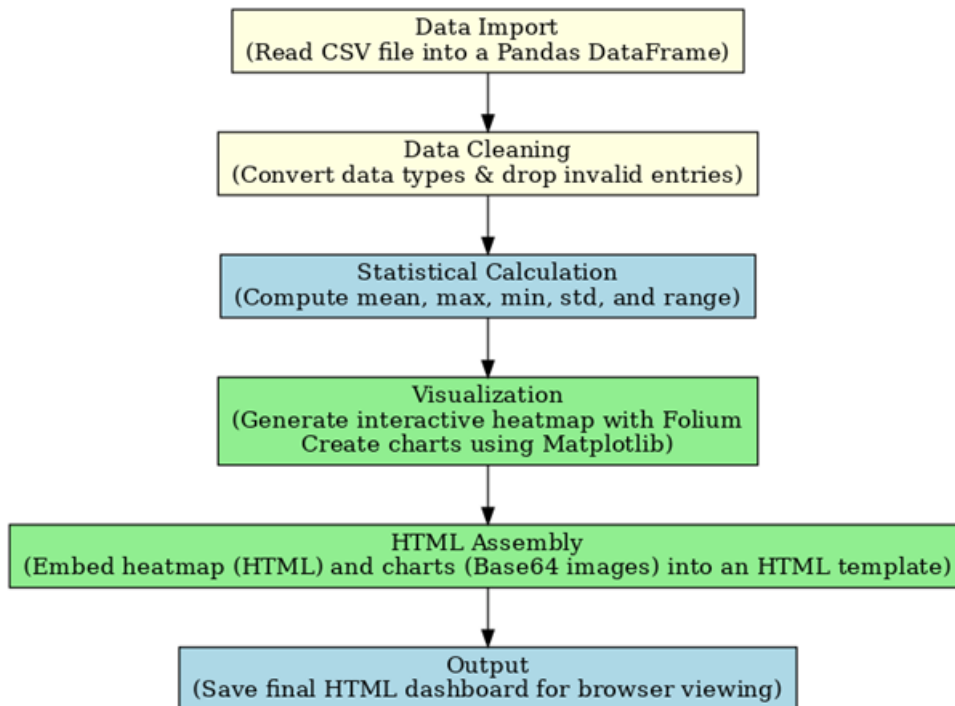


Figure 5.33:
Python
Flowchart

The data processing begins with importing the CSV file into a Pandas DataFrame. Once loaded, the data is cleaned by converting GPS coordinates and EMR values to numeric types and removing invalid entries.

Next, key statistical metrics including mean, max, min, standard deviation, and range are computed to analyze the dataset. The cleaned data is then visualized, starting with an

interactive heatmap generated using Folium, followed by additional charts such as a bar chart, line chart, and boxplot created with Matplotlib.

To assemble the final output, the heatmap (as an HTML element) and the charts (converted to Base64 images) are embedded into an HTML template. The final dashboard is then saved and can be viewed in a web browser.

5.5.4 Detailed Explanation of the HTML Dashboard

The final component of the Python module is the interactive HTML dashboard, which integrates the visualizations and statistical insights into a single, user-friendly interface.

HTML Template and Dynamic Embedding

The HTML template is structured to provide a modern, responsive design. It uses Tailwind CSS for styling and Lucide icons for visual enhancements. The template includes separate sections for the interactive map and the statistical charts.

Head Section

External libraries such as Tailwind CSS and Lucide are loaded to ensure a modern look and responsive design. The `<style>` block includes custom styling to improve the appearance of elements like images and statistic cards.

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>EMR Data Dashboard</title>
  <script src="https://cdn.tailwindcss.com"></script>
  <script src="https://unpkg.com/lucide@latest"></script>
  <style>
    body {{ font-family: system-ui, -apple-system, sans-serif; }}
    .hidden {{ display: none; }}
    .stat-card:hover {{
      transform: scale(1.05);
      transition: transform 0.3s ease;
      box-shadow: 0 10px 15px rgba(0, 0, 0, 0.1);
    }}
    img {{
      width: 100%;
      height: auto;
      border-radius: 0.5rem;
      box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
    }}
  </style>
</head>
```

Figure 5.34:
HTML-Head

Navigation and Tab System

The navigation bar at the top allows users to switch between the map view and the statistics view. A simple JavaScript function controls the display of these sections.

```
<nav class="bg-gray-800 p-4 shadow-md">
  <div class="container mx-auto flex justify-between items-center">
    <div class="flex items-center gap-2">
      <i data-lucide="map-pinned" class="text-blue-400 w-6 h-6"></i>
      <span class="text-lg font-semibold">EMR-Mapping </span>
    </div>
    <div class="flex gap-2">
      <button onclick="showTab('map')" id="mapBtn" class="
px-4 py-2 bg-gray-700 rounded-lg text-gray-300 flex items-center gap-2">
        <i data-lucide="map" class="w-4 h-4"></i> Map
      </button>
      <button onclick="showTab('stats')" id="statsBtn" class="
px-4 py-2 bg-gray-700 rounded-lg text-gray-300 flex items-center gap-2">
        <i data-lucide="bar-chart-3" class="w-4 h-4"></i> Statistics
      </button>
    </div>
  </div>
</nav>
```

Figure 5.35:
Navigation
Bar

Responsive Layout

The use of Tailwind CSS classes ensures that the dashboard is responsive and adapts well to different screen sizes.

```
<main class="container mx-auto p-6">
  <div id="map" class="tab">
    <h1 class="text-2xl font-bold mb-4">Heatmap</h1>
    <div class="bg-white rounded-lg shadow-md p-4">{map_html}</div>
  </div>
  <div id="stats" class="tab hidden">
    <h1 class="text-2xl font-bold mb-4">Statistics</h1>
    <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
      <div class="bg-white p-4 rounded-lg shadow-md flex items-center justify-between stat-card">
        <div>
          <p class="text-sm text-gray-500">Average EMR</p>
          <p class="text-xl font-bold text-gray-800">{average_emf:.2f} mV</p>
        </div>
        <i data-lucide="radio-tower" class="w-6 h-6 text-blue-500"></i>
      </div>
      <div class="bg-white p-4 rounded-lg shadow-md flex items-center justify-between stat-card">
        <div>
          <p class="text-sm text-gray-500">Max EMR</p>
          <p class="text-xl font-bold text-gray-800">{max_emf:.2f} mV</p>
        </div>
        <i data-lucide="arrow-up" class="w-6 h-6 text-green-500"></i>
      </div>
    </div>
  </div>
```

Figure 5.36:
Responsive
Layout

Visual Examples and Screenshots

To fully illustrate the functionality of the dashboard, I have included actual screenshots of the final product:

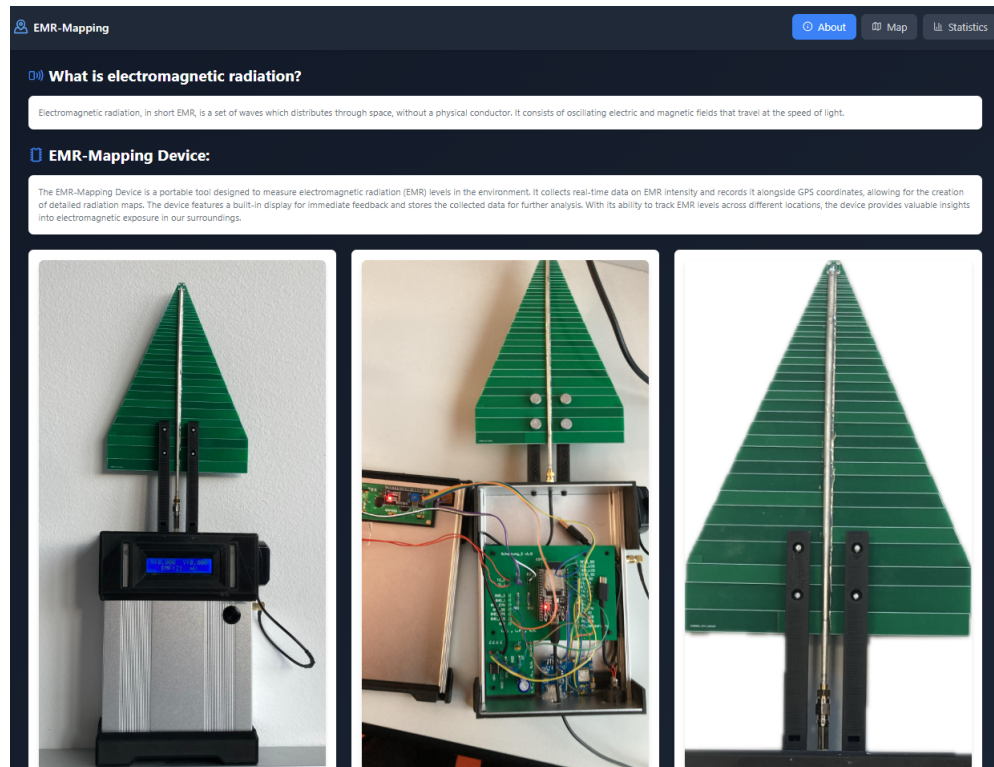


Figure 5.37:
About
View

This screenshot showcases the About page, which provides insights into EMR, details about the EMR-Device we created (including images), and introduces our team with photos and descriptions of our roles in the project (here you cannot see the team overview).

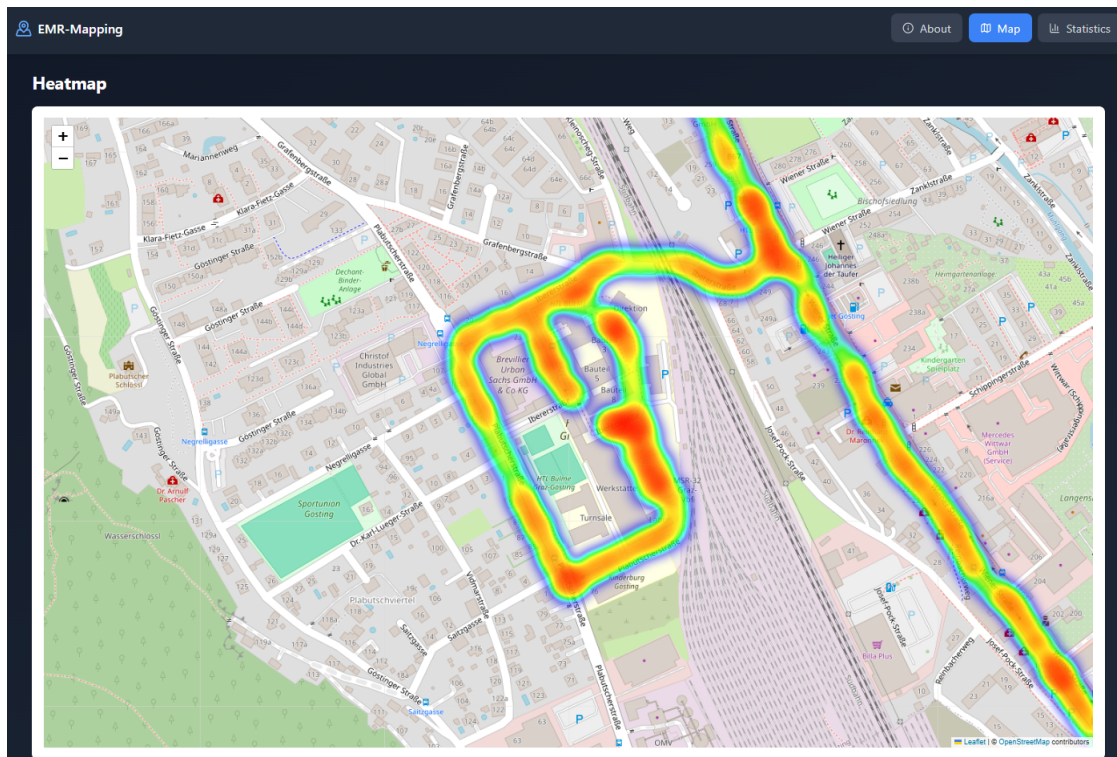


Figure 5.38:
Map
View

This screenshot displays the interactive heatmap, where the color gradient from green to red represents the intensity of EMR values. Users can zoom and drag the map to explore different areas.

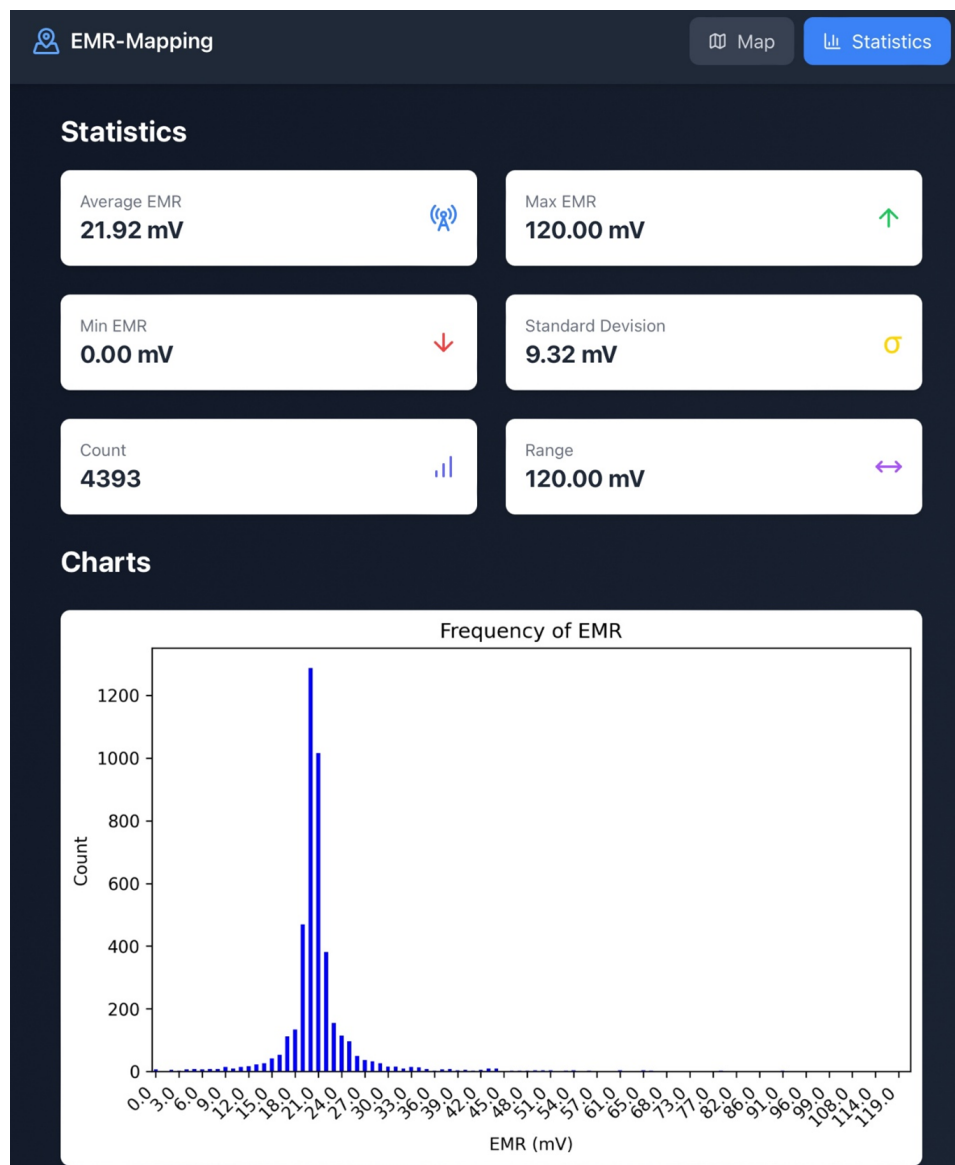


Figure 5.39:
Statistics
View

This screenshot shows the statistical charts (here you can only see the line chart) along with key metrics such as average, maximum, and minimum EMR values, providing an overview of the sensor data.

Literature Research

Letissa Adlboller

6.1 Introduction

All humans are exposed to electromagnetic fields (EMF) and electromagnetic radiation (EMR) from both natural and man-made sources. However, humans in modern day society are exposed to a growing number of man-made EMFs. Wireless technologies such as mobile phones and their base stations, personal computers (PC), cellular networks, Wi-Fi, radio transmission and much more have emerged over the last two decades, and the number of users has grown rapidly. This has raised a lot of concerns in the general public about potential health effects of electromagnetic radiation emitted by such devices and networks. People are concerned that electromagnetic radiation is dangerous and can cause cancer and many other diseases.

Nonetheless it is important to state that the exposure of non-ionizing radiation, which includes electromagnetic radiation, has to comply with the current exposure guidelines. These principles for protection have been developed by the International Commission on Non-Ionizing Radiation Protection (ICNIRP) [64] [65] [66] as well as the International Commission for Radiological Protection (ICRP) [67] . These Committee provide scientifically based advice and guidance on protection against adverse effects of non-ionizing radiation. Yet the general public is still doubtful about the safety standards. The problem is the lack of knowledge about this topic as the most convenient way to obtain knowledge is through the media such as TV or newspapers and not through scientific studies. People have a tough time distinguishing between the highly energetic ionizing radiation associated with atomic bombs and nuclear power plants and the much less energetic radiation, called non-ionizing radiation, emitted by mobile phones, base stations, Wi-Fi, radar and electrical power lines. This contributes to the safety concerns of the public.

However, these concerns have not just appeared over the last two decades. Already in the 1950s to 1960s, when the radar was invented, people were concerned whether it could have

health effects on humans or not. In the 1970s when AM (amplitude modulation), FM (frequency modulation) and TV (television) broadcasting antennas were firstly introduced, people were also concerned about possible health effects. Since the 1990s, most of the concerns have been about wireless technologies, due to the growing numbers of mobile phone users and therefore the growing number of mobile phone base stations, Wi-Fi Networks and other Technologies. This suggests that these concerns are not just modern issues but have existed for a long time [68].

Considering the following statements of the WHO concerning EMFs health effects, it can be concluded that the available scientific evidence we have is sufficient in proving not only that there are no potential health risks but also proving that our current safety standards are reliable: “Despite the feeling of some people that more research needs to be done, scientific knowledge in this area is now more extensive than for most chemicals.” as well as “. . . the WHO concluded that current evidence does not confirm the existence of any health consequences from exposure to low level electromagnetic fields.”[69].

It is, however, important to note that some gaps in research into health effects exist and that they need further investigation. We can't know for sure if our guidelines meet the protection standard we need. Especially because we lack long period studies (10>), which would give more information about the possible risk of cancer or other illnesses which normally take a longer time to develop than short term health effects. Furthermore, it is not excluded that the exposure of higher levels of electromagnetic fields could in fact cause related health problems.

6.2 Current Safety Standards

Like for other subjects there are also existing standards to limit overexposure from electromagnetic fields in order to protect the health of the public.

Each country sets their own national standard for the exposure to electromagnetic fields. Most of the national standards set by the countries rely on the guidelines set by the International Commission on Non-Ionizing Radiation Protection (ICNIRP) [65][66]. This organization is non-governmental and bases their guidelines on scientific evidence from all over the world. They review their guidelines frequently and update them if necessary.

The following table (Table 1) represents a summary of the ICNIRP guidelines set for the public exposure as well as for the occupational exposure.

Summary of the ICNIRP exposure guidelines [65]

	European power frequency		MPBS frequency		Microwave oven frequency	Mobile Phones
Frequency range	50Hz	50Hz	900Hz	1.8GHz	2.45GHz	900MHz - 18000MHz
	Electric field (V/m)	Magnetic field (uT)	Power density (W/m ²)	Power density (W/m ²)	Power density (W/m ²)	Specific energy absorption rate* (W/kg-1)
Public exposure	5000	100	4.5	9	10	2
Occupational exposure	10000	500	22.5	45	/	10

Occupational VS Public exposure limits

The limits differ between the general public exposure and occupational population exposure. Adults who generally experience electromagnetic fields in occupational settings are trained to be aware of the potential risks and to take precautions to minimize their exposure. As for the general public, individuals are often unaware of their exposure to EMF and cannot be expected to take actions in order to minimize their exposure.

Typical public exposure

The following table (Table 2) presented by the WHO Regional Office for Europe, could help to better visualize typical at home and environmental exposure, in contrast to the current safety standards.[69]

Source	Typical maximum public exposure	
	Electric field (V/m)	Magnetic flux density (uT)
Natural fields	200	70 (earth 's magnetic field)
Mains power (in homes not close to power lines)	100	0.2
Mains power (beneath large power lines)	10000	20
Electric trains and trams	300	50
TV and computer screens	10	0.7
Typical maximum public exposure (W/m2)		
TV and radio transmitters		0.1
Mobile phone base stations		0.1
Radars		0.2
Microwave ovens		0.5

Exposure above guidelines

The overall public does not experience electromagnetic fields above the safety limits, the typical exposure is far below these values. Electromagnetic exposure within the safety limits is ensured to cause no known adverse health effects. Even after experiencing exposure several times higher than the given limit, the exposure would still be within the safety margin, due to the large safety factors which are incorporated to the level known to cause a health consequence [69].

6.3 Wireless Technologies and related Health effects

6.3.1 Mobile phone base stations (MPBS)



Figure 6.1:
Mobile phone base
station [70]

With the increasing demand for wireless communication, mobile phone base stations play a vital role in maintaining a stable connection between mobile devices and the broader telecommunication network. As this technology continues to evolve -from 2G and 3G to 4G LTE and now 5G- mobile phone base stations have advanced in higher data speed, lower latency and increased capacity. The widespread use of this technology has increased the number of MPBS all over the world. And has therefore led to public concerns about the potential health effects of exposure to radiofrequency electromagnetic fields from mobile phone base stations [71] [72] [73]. Including potential health impacts such as headaches, sleeping disorders, cognitive effects or even cancer and adverse birth outcomes.

The findings on health effects concerning the EMF emitted by MPBS have been inconsistent and ambiguous. There are various crucial points that must be considered when researching in this field and conducting studies. Firstly, the exposure of the MPBS which is defined by intensity, duration and pattern, making it challenging to measure it accurately. Secondly the study type which is used to perform the study. When using an unblinded study type, the risk for unreliable results is rather high, as the reported symptoms could be influenced by psychological factors such as the fear of EMF exposure [74]. Therefore, the inconsistent results of various studies can be attributed to the variation in approach and design of the different studies, and in the different criteria on which the assessment was based.

Self-reported symptoms (sleep quality, headaches, etc....)

Unblinded Studies

An unblinded cross sectional study, randomly selected inhabitants who lived near 10 selected base stations, and performed cognitive tests and evaluated information about their wellbeing as well as their sleep quality in correlation to the EMF exposure emitted by these base stations. The findings of this study show that some symptoms, especially headaches, show a significant relation to the measured power density. Moreover, with increasing exposure levels, the perceptual speed of the participants increased, while accuracy decreased. In terms of sleep quality, they found that exposure had no significant effect [75]. However, only when considering the participants' fears or concerns about the possible negative health effects of base stations. This suggests that there is a high chance that these symptoms were not caused by the actual EMF exposure itself, but rather by fear and anxiety about possible negative effects caused by the exposure.

Another unblinded cross-sectional study in Egypt, discovered that inhabitants who lived nearby the mobile phone base station are at risk of developing neuropsychiatric problems and changes in performance during neurobehavioral tests [76].

Blinded Studies

Very different results were found in blind study designs, where participants couldn't know what kind of measure they were exposed to. Participants seemed rather unaffected by the EMF exposure. In a German experimental double-blind study, a newly installed base station (UMTS)

nearby an office was turned on or off for a period of three months. 95 employees of this office building in Bavaria (Germany) were then asked to report their health status on every working day twice a day, before their work started and after their work ended. They were asked to fill out a questionnaire consisting of 21 Symptoms they could choose from. In addition, they were also asked whether they felt like the base station was turned on or off that day. The study found no correlation between the self-reported symptoms of the employees and the state of the base station. They did, however, notice that when the employees thought that the base station was in use, they reported more health complaints and the other way around [77].

A randomized double-blind provocation study from 2010, investigated the health effects of short-term exposure to a TETRA (Terrestrial Trunked Radio) base station in a semianechoic chamber. Participants with self-reported "electro sensitivity" and of participants who served as controls participated in an open provocation test. Multiple scales and measures were used to evaluate participants well-being, offering valuable insights. They found no difference in any measure between the real TETRA and the sham exposure, for both control and electrosensitive groups. However, under unblinded conditions, participants with self-diagnosed electro sensitivity, reported feeling worse under TETRA exposure than under Sham exposure. The conclusion: "Our findings suggest that the adverse symptoms experienced by electrosensitive individuals are due to the belief of harm from TETRA base stations rather than to the low-level EMF exposure itself." [78]

This Comparison demonstrates, how psychological factors play a huge role in the outcome of such studies. Whether participants know or don't know that they are exposed to a source of EMF makes a difference in how they perceive their own well-being. Especially when there is fear and concerns about possible harm from EMF exposure. This provides strong evidence that the outcome of the study is dependent on the study-type that is used.

6.3.2 Mobile Phones



Figure 6.2:
Mobile phone [79]

The mobile phone is an important invention in human history, which has revolutionized people's lifestyle. It makes everyday life a lot easier by allowing us to use it in various ways. It allows us to:

- have unlimited access to information about various subjects
- being available all the time through text messages, phone calls, video calls and social media
- take pictures and videos in good quality anywhere
- have access to music, games, entertaining content and much more

That's what makes it so convenient, and also easy to carry with us everywhere we go, due to its compact design. Which is also why people are concerned that the constant exposure of electromagnetic radiation emitted by mobile phones could be hazardous to us.

The controversy about possible health effects emitted by mobile phones is huge. On the one hand, big Commissions such as the International Commission on Non-Ionizing Radiation Protection (ICNIRP) [64] as well as the International Commission for Radiological Protection (ICRP) [67] and the World Health Organization (WHO) [69] claim that electromagnetic radiation causes no health effects other than the heating of body tissues. On the other hand, the International Agency for Research on Cancer (IARC) classified radiofrequency electromagnetic fields as possibly carcinogenic to humans (Group 2B) [80]. Many other studies use this classification to support their arguments on the possibility of electromagnetic radiation causing cancer [81] [82] [83].

While mentioning several other health effects such as sperm damage, DNA integrity, immune disfunction, thyroid gland disorders, diabetes, Alzheimer, Parkinson disease and much more.

Cancer

A Group 2B cancer classification refers to the title as “possible carcinogenic to humans”. The evidence in this classification is described as limited and inadequate, agents in this classification therefore have limited evidence of carcinogenicity in humans and less than sufficient evidence of carcinogenicity in experimental animals. The IARC Director Christopher Wild said: "Given the potential consequences for public health of this classification and findings, it is important that additional research be conducted into the long- term, heavy use of mobile phones” [80].

This classification is heavily based on the interview-based case-control study called the Interphone study [84], which was at that time the largest of its kind. The Study was made available for the working group shortly before the classification was due to be summoned. The Interphone study investigated the risk of glioma or meningioma in correlation to mobile phone use. They concluded that they didn't observe an increased risk for glioma or meningioma after ten years of first mobile phone use. However, they suggested that there are indications of possible risk of glioma and meningioma in heavy phone users.

The authors of this study, however, admitted that due to several biases and errors, no clear conclusion can be drawn from this study. Several comments and reviews can be found, commenting on the Results of the Interphone study. Mentioning how the study has methodological deficits as well as several biases, which limit the interpretation of its findings and make it doubtful of detecting a correlation between phone usage and brain tumors [85] [86] [87]. Given the lack of proper studies available on this subject, the reviews and comments conclude their findings as uncertain. They call for further research requesting cohort studies in order to investigate adult brain cancers. Noting the importance of the time duration for the study, which should be at least above ten years as well as mentioning the importance of minimizing bias as good as possible.

When considering the classification about the possible carcinogenicity from electromagnetic radiation emitted by mobile phones, it may be helpful to see it from a different angle. Especially this topic seems to get a lot of attention while other surprisingly common or normalized things which also fall under this classification, get way less attention, for instance: In the year 2013 the IARC classified Aloe Vera (whole leaf extract) as possibly carcinogenic (Group 2B) [88], pickled vegetables (traditional Asian) are also possibly carcinogenic to humans (Group 2B) [89]. These substances are also very common in our everyday life and seem to be consumed without any consequences.

Male Infertility

There has been a significant increase in male infertility over last few decades. The data suggest that this world-wide incline continues in the 21st century at an accelerated pace [90]. It is known that male infertility is attributed to environmental as well as several lifestyle factors such as:

smoking cigarettes, alcohol intake, use of illegal drugs, obesity, psychological stress, advanced paternal age, eating habits as well as coffee consumption [91][92][93] [94].

Considering the rise of mobile phone use over the last few decades and the increase of male infertility at the same time, studies suggest that male infertility is also likely to be caused by non-ionizing radiation emitted by cell phones or laptops, when kept near the scrotum [95][96][97]. The amount of radiofrequency energy absorbed by human tissue is measured by the specific absorption rate (SAR) which is legally limited by the International Commission on non-ionizing Radiation Protection (ICNIRP) to 2.0 W/kg (localized exposure: head and trunk) [95]. SAR values for mobile phones usually range from 0.12 to 1.6 W/kg body weight, depending on the model of the phone [96]. However, authors insist that many factors affect the amount of radiation which is absorbed by humans such as the duration of exposure, the mode of usage of the device (standby mode versus making a phone call) as well as the position of the phone (near head, in pocket, on the lap) [98]. Studies claim that the radiation affects the quality of the semen by decreasing the sperm count, mobility, viability and morphology, which could contribute to male infertility [99] [100].

Human Studies

An observational study investigated the effect of cell phone use on the semen quality of three hundred sixty-one men who attended an infertility clinic. According to their active cell phone use they were divided into four groups: group A: no use; group B: <2h/day; group C: 2-4h/day; group D: >4h/day. The study claims to have found a decrease in sperm count, mobility, viability, and normal morphology with the increase in daily use of cell phones. However, the level of radiation exposure was based on self-reporting, also the study did not mention for how long the participants were exposed to cell phone radiation or how (e.g. phone in pocket). Moreover, the study did not take environmental factors, eating habits, stress or occupational exposure into account [99].

One study claimed that using cell phones for more than 10 years was positively related to the percentage of atypical sperm and the percentage of sperm head abnormalities and was also negatively associated with the percentage of motile sperm cells. The study consisted of three hundred forty-four men. Their cell phone use was based on the years of usage of this equipment (0–5 years, 6–10 years, and 11–25 years). In addition, participants gave information about consumption of alcohol, tobacco, coffee intake, wearing loose underwear, and cell phone and sauna usage [101].

This study was included in a recent meta analyses which reviewed the reliability of the evidence of several studies concerning the effects of radiofrequency exposure on male fertility [102]. The meta Analysis reviewed that all included studies were rated as probably or definitely at high risk of bias for at least two domains on the OHAT risk of bias tool, including [101] which showed a definitely high risk of exposure characterization bias as well as a probably high risk for selective reporting bias. Not only was the exposures characterization across studies often inconsistent and

lacking in detail, but also Specific Absorption Rate (SAR) measurements were not included in the exposure parameter. The studies in general were -due to several risks of bias, unexplained inconsistency, indirectness and imprecision- rated as very low-certainty evidence. Therefore, the authors concluded the current evidence, regarding the effects of RF-EMF from mobile phones on sperm concentration, morphology, progressive motility and total sperm count in the general public, as uncertain.

In-Vitro-Studies

Another study examined the sperm quality of 32 healthy men with the influence of direct mobile radiation. Sperm samples were taken and divided into two equal samples A and B, both samples were put into a thermostat. Additionally in the thermostat of sample B, a mobile phone in standby/talk mode was placed for the duration of 5 hours. The study affirms to have found a correlation between mobile phone radiation exposure, DNA-fragmentation level and decreased sperm motility [103]. Nonetheless, the sample of this study was rather small which decreases the statistical significance. Although the men were described as healthy, other factors such as stress, eating habits, smoking, alcohol consumption – which are known to have an impact on the semen quality () - were not taken into consideration. Furthermore, the in-vitro-setting could differ from realistic exposure, as the radiation is not directly exposed to the semen.

The main problem with in vitro studies, concerning this topic is the fact that the absorbed electromagnetic radiation from the samples cause a temperature rise, depending on exposure time and radiation strength. In order to distinguish thermal effects from non-thermal effects, it's crucial to measure the temperature of the samples before and after the exposure. In addition, a negligible temperature difference between exposed and sham-exposed sample must be ensured. Another difficulty in vitro studies is the estimation from SAR levels which need special computers as well as deep knowledge about the topic in order to be measured and calculated rightfully [104].

Based on the study mentioned above [103], it's obvious that these problems occurred. The study mentioned placing the sperm samples in thermostats at controlled temperatures. However, the study did not explicitly mention active temperature monitoring during exposure or a controlled cooling process to minimize thermal differences. In addition, the study does not mention the SAR value at which the sample was exposed to, nor did the study consider different SAR distribution all over the samples which could have led to hot spots where exposure could have led to extreme thermal effects on some spots on the sample.

Due to several problems that occurred in the study it's possible that the results were confused with simple thermal effects, which would make the correlation between electromagnetic radiation and decreasing sperm quality insufficient.

6.4 Conclusion

Considering all the data and studies we have about electromagnetic radiation in correlation with health effects such as self-reported symptoms (headaches, sleep disturbance), cancer (head tumors: glioma and meningioma) as well as male infertility and several others, it can be said that experts can't agree on one clear answer. The question of whether electromagnetic radiation is harmful to us or not will remain unsolved. At least for as long poorly controlled studies about this topic will continue being conducted and published.

Further data must be researched and conducted with a better approach and structure in particular: longer study time (10 > years), more accurate measuring of exposure, minimization of bias as good as possible, considering possible other factors which could contribute to certain results (thermal effects, environmental factors, lifestyle habits, etc.) and in general appropriate statistical interpretation of results. This would build a greater base of evidence and give results better reliability and certainty in determining any possible EMF-effects.

Even though there remains some uncertainty about how harmful electromagnetic exposure really is to us, the evidence leans towards the harmless side. The only scientifically assured effect from EMF is the heating from human tissue due to absorption from electromagnetic radiation, the phones battery and the electric circuit during operation. To protect the public from any possible health effects coming from exposure to electromagnetic radiation, organizations such as ICNIRP as well as the WHO and ICRP frequently review their safety limits and update these standards to the latest scientific knowledge.

The European Union already implemented these standards into their policy in 1999. However, countries within the European Union handle these standards differently. Austria for instance has implemented these standards into their policy: Standard ÖNORM 1119 covers 0 Hz-30 kHz and standard ÖNORM 1120 covers 30 kHz-3000 GHz [105].

While more research is being conducted the WHO suggested some actions that could be taken in the meantime [69] :

More countries such as these in Europe should implement the ICNIRP standards into their policy as they protect the public with a large safety factor from any harmful effects. Areas where the electromagnetic exposure may reach values above the given limits, should be protected with a barrier to protect them from unauthorized access. During the planning of new power lines or mobile phone base stations it could help to openly communicate with the public to create better understanding and acceptance of new facilities. Lastly, to reduce any mistrust and fear about EMF it could help to openly communicate with the public about health information and with scientists to raise general awareness about this topic.

List of acronyms

AM		Amplitude Modulation
ASA		Acrylic Styrene Acrylonitrile
BOM		Bill Of Materials
CAD		Computer-Aided Design
CE		Conducted Emission
DRC		Design Rule Check
EMC		Electromagnetic Compatibility
EMF		Electromagnetic Fields
EMI		Electromagnetic Interference
EMR		Electromagnetic Radiation
FM		Frequency Modulation
GPS		Global Positioning System
IARC		International Agency for Research on Cancer
ICNIRP		International Commission On Non-Ionising Radiation Protection
ICRP		International Commission for Radiological Protection
LCD		Liquid Crystal Display
MPBS		Mobile Phone Base Station
OLED		Organic Light Emitting Diodes
op-amp		Operational Amplifier
PLA		Polylactic Acid
RE		Radiated Emission
RF		Radio Frequency
SAR		Specific Absorption Rate
TETRA		Terrestrial Trunked Radio
UMTS		Universal Mobile Telecommunications System
WHO		World Health Organisation
Wi-Fi		Wireless Local Area Network

APPENDIX B

Appendix

B.1 3D printed parts

Lisa Kolli

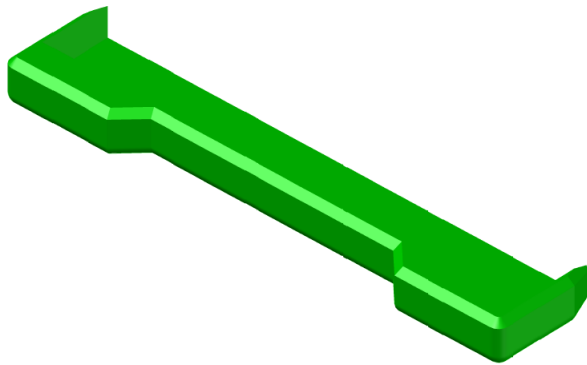


Figure B.1:
3D part in
black on the
back of the
device

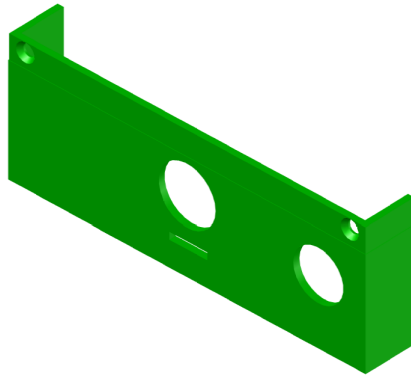


Figure B.2:
3D part in black on
the bottom of the
device

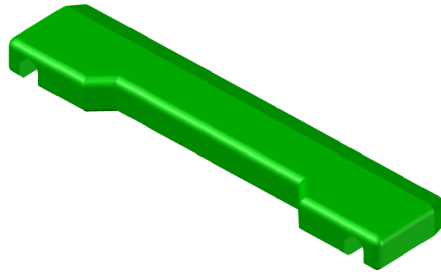


Figure B.3:
3D part in black on the
front of the device

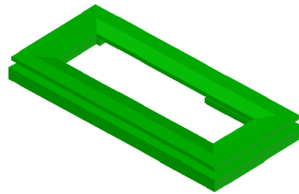


Figure B.4:
two 3D parts in black
for the LC Display

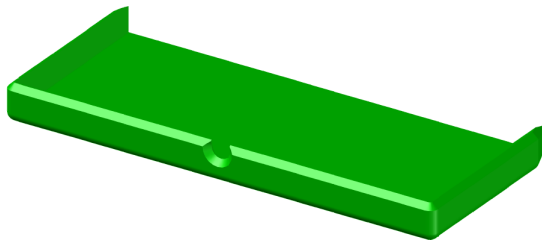


Figure B.5:
3D part in black
on the back of
the device

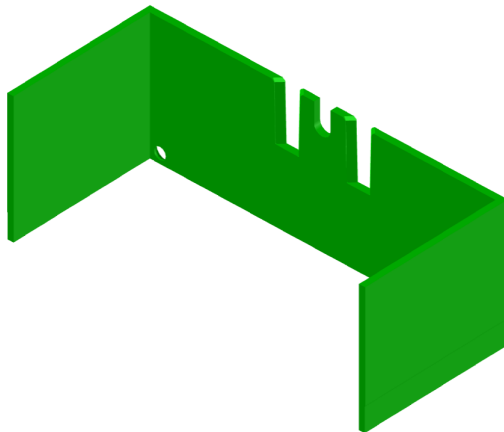


Figure B.6:
3D part in
black on the
top of the
device

B.2 Complete code

Robin Heindl

B.2.1 ESP32 firmware code

```

1  #include <Wire.h>
2  #include <LiquidCrystal_I2C.h>
3  #include <SdFat.h>
4  #include <SPI.h>
5  #include <TinyGPS++.h>
6
7  // Defines RX and TX pins for Serial 2 + GPS baud rate
8  #define RXD2 16
9  #define TXD2 17
10 #define GPS_BAUD 9600
11
12 // Pin configuration
13 const int chipSelect = 5;           // SD card chip-select pin
14 const int emfPin = 4;              // Analog pin for the EMF sensor
15 const int greenLEDPin = 32;        // Green LED for "Loading"
16 const int redLEDPin = 2;           // Red LED for "GPS signal received"
17 const int buttonPin = 33;          // Button to start/stop saving
18 bool isLoading = false;            // Variable to store loading state
19 unsigned long outputTime = 0;      // To delay LCD and serial output
20 unsigned long outputDelay = 2000;  // Output delay time
21 unsigned long gpsSignalTime = 0;   // Time of last GPS signal
22 unsigned long lastSaveTime = 0;    // Time of last save
23 unsigned long saveInterval = 1000; // Interval of 1 second (1000 ms)
24 const unsigned long gpsLedDuration = 500; // LED blink duration in ms
25 bool previousButtonState = false;  // Tracks previous button state
26
27 // LCD setup (I2C address: 0x27, 16 columns, 2 rows)
28 LiquidCrystal_I2C lcd(0x27, 16, 2);
29
30 // Create SdFat object
31 SdFat SD;
32 File dataFile;
33
34 // TinyGPS++ object
35 TinyGPSPlus gps;
36
37 // Create an instance of the HardwareSerial class for Serial 2
38 HardwareSerial gpsSerial(2);
39
40 // Variables for storing values
41 float xValue;
42 float yValue;
43 int emfValue; // As an integer in mV
44
45 // Declare functions
46 void setupLCD();
47 void setupSDCard();
48 void writeCSVHeader();
49 void displayValues(float x, float y, int emf);
50 void saveToCSV(float x, float y, int emf);
51 int readEMF();
52 void checkButton();
53 void GPS();

```

```

55 void setup() {
56     // Serial communication for debugging
57     Serial.begin(115200);
58
59     // Initialize LCD and SD card
60     setupLCD();
61     setupSDCard();
62
63     // Write CSV header
64     writeCSVHeader();
65
66     // Configure LED pins as output
67     pinMode(greenLEDPin, OUTPUT);
68     pinMode(redLEDPin, OUTPUT);
69
70     // Configure button as input with pull-up resistor
71     pinMode(buttonPin, INPUT_PULLUP);
72
73     // Initialize loading state
74     digitalWrite(redLEDPin, LOW); // Start with "Not loading"
75     digitalWrite(greenLEDPin, LOW);
76
77     // Start Serial 2 with defined RX & TX pins and baud rate
78     gpsSerial.begin(GPS_BAUD, SERIAL_8N1, RXD2, TXD2);
79     Serial.println("Serial 2 started at 9600 baud rate");
80
81     delay(2000); // Wait to show display
82     lcd.clear();
83 }
84
85 void loop() {
86     // Check button state and toggle loading
87     checkButton();
88
89     // Retrieve GPS values
90     GPS();
91
92     // Read EMF value from the sensor on analog pin 4
93     emfValue = readEMF();
94
95     int Millis = millis();
96     if ((Millis - outputTime) > outputDelay) {
97         // Display values on LCD and serial monitor
98         displayValues(xValue, yValue, emfValue);
99
100        Serial.print("X: ");
101        Serial.print(xValue, 2);
102        Serial.print(", Y: ");
103        Serial.print(yValue, 2);
104        Serial.print(", EMF: ");
105        Serial.print(emfValue); // EMF value without decimals
106        Serial.println(" mV");
107
108        outputTime = millis();
109    }

```

```

111     if (isLoading) {
112         // Green LED lights up while loading
113         digitalWrite(greenLEDPin, HIGH);
114         if (gpsSerial.available() > 0) {
115             gps.encode(gpsSerial.read());
116             // Check if GPS data is valid and enough time has passed si
nce last save
117             if
(isValidGPS() && (millis() - lastSaveTime >= saveInterval)) {
118                 if (xValue > 0 && yValue > 0) {
119                     // Save values to CSV file
120                     saveToCSV(xValue, yValue, emfValue);
121
122                     // Update last save time
123                     lastSaveTime = millis();
124                 }
125             }
126         }
127     } else {
128         // Green LED turns off when not loading
129         digitalWrite(greenLEDPin, LOW);
130     }
131 }
132
133 bool isValidGPS() {
134     // Checks if GPS coordinates are fresh and valid
135     return (gps.location.isUpdated() && gps.location.isValid());
136 }
137
138 // Initialize LCD
139 void setupLCD() {
140     lcd.init();
141     lcd.backlight();
142     lcd.setCursor(0, 0);
143     lcd.print("EMF Meter Start");
144 }
145
146 // Initialize SD card
147 void setupSDCard() {
148     if (!SD.begin(chipSelect, SPI_HALF_SPEED)) {
149         Serial.println("Error: SD card not recognized");
150         lcd.setCursor(0, 1);
151         lcd.print("SD Init Error");
152         while (!SD.begin(chipSelect, SPI_HALF_SPEED))
153             ;
154     }
155
156     Serial.println("SD card successfully initialized.");
157     lcd.setCursor(0, 1);
158     lcd.print("SD Init OK");
159 }

```

```
161 // Write CSV header
162 void writeCSVHeader() {
163     dataFile = SD.open("data.csv", FILE_WRITE);
164     if (dataFile) {
165         dataFile.println("X,Y,EMF");
166         dataFile.close();
167         Serial.println("CSV file created.");
168     } else {
169         Serial.println("Error: Could not open CSV file.");
170     }
171 }
172
173 // Display values on LCD (X & Y with 2 decimals, EMF as integer)
174 void displayValues(float x, float y, int emf) {
175     lcd.setCursor(0, 0);
176     lcd.print("X:");
177     lcd.print(x, 3);
178     lcd.setCursor(9, 0);
179     lcd.print("Y:");
180     lcd.print(y, 3);
181     lcd.setCursor(3, 1);
182     lcd.print("EMF:");
183     lcd.print(emf);
184     lcd.setCursor(10, 1);
185     lcd.print(" mV");
186 }
187
188 // Save values to CSV
189 void saveToCSV(float x, float y, int emf) {
190     dataFile = SD.open("data.csv", FILE_WRITE);
191     if (dataFile) {
192         dataFile.print(x, 6);
193         dataFile.print(",");
194         dataFile.print(y, 6);
195         dataFile.print(",");
196         dataFile.println(emf);
197         dataFile.close();
198         Serial.println("Data saved to CSV:");
199         Serial.print(x);
200         Serial.print(",");
201         Serial.print(y);
202         Serial.print(",");
203         Serial.println(emf);
204     } else {
205         Serial.println("Error: Could not open CSV file.");
206     }
207 }
208
```

```
209 // Read EMF value and convert to millivolts
210 int readEMF() {
211     int sensorValue = analogRead(emfPin);
212     float voltage = sensorValue * (3.3 / 4095);
213     return (int)((voltage / 23) * 1000);
214 }
215
216 // Check button state with debounce
217 void checkButton() {
218     bool currentState = !digitalRead(buttonPin);
219     // Read button (inverted due to INPUT_PULLUP)
220
221     if (!currentState && previousButtonState) {
222         // Change state only when button is released
223         isLoading = !isLoading;
224     }
225
226     previousButtonState = currentState;
227 }
228
229 void GPS() {
230     // Read GPS data
231     while (gpsSerial.available() > 0) {
232         gps.encode(gpsSerial.read());
233         if (gps.location.isValid()) {
234             gpsSignalTime = millis();
235         }
236     }
237
238     if (gps.location.age() > 2000) {
239         digitalWrite(redLEDPin, LOW); // LED off
240     } else {
241         unsigned long currentMillis = millis();
242         if ((currentMillis - gpsSignalTime) % (2
243             * gpsLedDuration) < gpsLedDuration) {
244             digitalWrite(redLEDPin, HIGH); // LED on
245         } else {
246             digitalWrite(redLEDPin, LOW); // LED off
247         }
248     }
249
250     if (gps.location.isUpdated()) {
251         xValue = gps.location.lng();
252         yValue = gps.location.lat();
253     }
```

B.2.2 Python data processing and visualization code

```

1  import os
2  import pandas as pd
3  import folium
4  from folium.plugins import HeatMap
5  from branca.colormap import LinearColormap
6  import matplotlib.pyplot as plt
7  import io
8  import base64
9  import numpy as np
10
11 # Define paths based on current directory
12 current_directory = os.path.dirname(os.path.abspath(__file__))
13 csv_path = os.path.join(current_directory, "data.csv")
14
15 # Check if CSV file exists
16 if not os.path.exists(csv_path):
17     raise FileNotFoundError(f"The file 'data.csv' was not found in {current_directory}.")
18
19 # Load CSV data into a DataFrame and clean the data
20 df = pd.read_csv(csv_path, header=0)
21 df.columns = ['Longitude', 'Latitude', 'EMR']
22 df['Latitude'] = pd.to_numeric(df['Latitude'], errors='coerce')
23 df['Longitude'] = pd.to_numeric(df['Longitude'], errors='coerce')
24 df['EMR'] = pd.to_numeric(df['EMR'], errors='coerce')
25 df = df.dropna(subset=['Latitude', 'Longitude', 'EMR'])
26
27
28 # Statistics
29 average_emf = df['EMR'].mean()
30 max_emf = df['EMR'].max()
31 min_emf = df['EMR'].min()
32 count_emf = len(df['EMR'])
33 range_emf = max_emf - min_emf
34 std_deviation = df['EMR'].std()
35
36 # Scale EMR values for visualization
37 df['EMR_Scaled'] = df['EMR'] / df['EMR'].max()
38
39 # Create a map centered around the average GPS coordinates
40 mymap = folium.Map(location=[df['Latitude'].mean(), df['Longitude'].mean()], zoom_start=12)
41
42 # Prepare heatmap data as a list of [latitude, longitude, scaled EMR]
43 heat_data = [[row['Latitude'], row['Longitude'], row['EMR_Scaled']] for index, row in
44               df.iterrows()]
45
46 # Generate the heatmap and add it to the map
47 HeatMap(heat_data, radius=25, blur=15, max_zoom=1).add_to(mymap)
48
49 # Add a linear color map (legend) to the map
50 colormap = LinearColormap(colors=['green', 'yellow', 'red'], vmin=0, vmax=144)
51 colormap.add_to(mymap)
52
53 # Store the HTML representation of the map
54 map_html = mymap._repr_html_()

```

```

56 # Function to convert Matplotlib figures into Base64 strings
57 def fig_to_base64(fig):
58     buf = io.BytesIO()
59     fig.savefig(buf, format='png', dpi=300, bbox_inches='tight')
60     buf.seek(0)
61     encoded = base64.b64encode(buf.read()).decode('utf-8')
62     buf.close()
63     return f"data:image/png;base64,{encoded}"
64
65 # Create charts and convert them to Base64
66 # Bar Chart
67 fig, ax = plt.subplots(figsize=(8, 4.5))
68 df['EMR'].value_counts().sort_index().plot(kind='bar', ax=ax, color='blue')
69 ax.set_title('Frequency of EMR')
70 ax.set_xlabel('EMR (mV)')
71 ax.set_ylabel('Count')
72 ax.set_xticks(ax.get_xticks()[::3])
73 ax.set_xticklabels(ax.get_xticklabels(), rotation=45, ha='right', )
74 bar_chart_base64 = fig_to_base64(fig)
75 plt.close(fig)
76
77 # Line Chart
78 fig, ax = plt.subplots(figsize=(8, 4.5))
79 ax.plot(df.index, df['EMR'], color='blue')
80 ax.set_title('EMR by Index')
81 ax.set_xlabel('Index')
82 ax.set_ylabel('EMR (mV)')
83 line_chart_base64 = fig_to_base64(fig)
84 plt.close(fig)
85
86 # Boxplot
87 fig, ax = plt.subplots(figsize=(8, 4.5))
88 ax.boxplot(df['EMR'], vert=False, medianprops=dict(color='blue'), showfliers=False, whis=[0, 100
89 ])
89 ax.set_xticks(np.round(np.linspace(min_emf, max_emf, 20)))
90 ax.set_xticklabels(ax.get_xticklabels(), rotation=45, ha='right', )
91 ax.set_title('Boxplot of EMR')
92 ax.set_xlabel('EMR (mV)')
93 boxplot_base64 = fig_to_base64(fig)
94 plt.close(fig)

```

```
96 # HTML-Template
97 html_template = f"""
98 <!DOCTYPE html>
99 <html lang="en">
100 <head>
101     <meta charset="UTF-8">
102     <meta name="viewport" content="width=device-width, initial-scale=1.0">
103     <title>EMR Data Dashboard</title>
104     <script src="https://cdn.tailwindcss.com"></script>
105     <script src="https://unpkg.com/lucide@latest"></script>
106     <style>
107         body {{ font-family: system-ui, -apple-system, sans-serif; }}
108         .hidden {{ display: none; }}
109         .stat-card:hover {{
110             transform: scale(1.05);
111             transition: transform 0.3s ease;
112             box-shadow: 0 10px 15px rgba(0, 0, 0, 0.1);
113         }}
114         img {{
115             width: 100%;
116             height: auto;
117             border-radius: 0.5rem;
118             box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
119         }}
120         .card {{
121             width: 100%;
122             max-width: 300px;
123             height: auto;
124             background-color: white;
125             border-radius: 0.5rem;
126             box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
127             padding: 1rem;
128             display: flex;
129             flex-direction: column;
130             align-items: center;
131             text-align: center;
132             overflow: hidden;
133             position: relative;
134         }}
```

```
136 .remove-when-use {{
137     text-align: center;
138     width: 100%;
139     position: absolute;
140     color: black;
141     top: 85px;
142     font-weight: bold;
143 }}
144
145 .details > p {{
146     font-size: .8em;
147     margin-top: .5em;
148 }}
149
150 .details > label {{
151     font-weight: bold;
152 }}
153
154 .details {{
155     color: white;
156     box-shadow: 0 3px 10px rgb(0 0 0 / 0.2);
157     width: 100%;
158     background-color: rgba(0,0,0,0.5);
159     position: absolute;
160     bottom: 0;
161     height: 2.6em;
162     transition: height .5s ease-in-out;
163     padding: 0.6em;
164     overflow: hidden;
165     left: 0;
166 }}
167
168 .card:hover > .details {{
169     height: 6.7em;
170 }}
171
172 .team-container {{
173     display: flex;
174     justify-content: center;
175     gap: 20px;
176     flex-wrap: wrap;
177 }}
```

```

179     </style>
180 </head>
181 <body class="bg-gradient-to-br from-gray-900 to-gray-800 text-white min-h-screen">
182     <nav class="bg-gray-800 p-4 shadow-md">
183         <div class="container mx-auto flex justify-between items-center">
184             <div class="flex items-center gap-2">
185                 <i data-lucide="map-pinned" class="text-blue-400 w-6 h-6"></i>
186                 <span class="text-lg font-semibold">EMR-Mapping </span>
187             </div>
188             <div class="flex gap-2">
189                 <button onclick="showTab('About')" id="AboutBtn" class="px-4 py-2 bg-gray-700 rounded-lg
text-gray-300 flex items-center gap-2">
190                     <i data-lucide="info" class="w-4 h-4"></i> About
191                 </button>
192                 <button onclick="showTab('map')" id="mapBtn" class="px-4 py-2 bg-gray-700 rounded-lg text
-gray-300 flex items-center gap-2">
193                     <i data-lucide="map" class="w-4 h-4"></i> Map
194                 </button>
195                 <button onclick="showTab('stats')" id="statsBtn" class="px-4 py-2 bg-gray-700 rounded-lg
text-gray-300 flex items-center gap-2">
196                     <i data-lucide="bar-chart-3" class="w-4 h-4"></i> Statistics
197                 </button>
198             </div>
199         </div>
200     </nav>
201     <main class="container mx-auto p-6">
202         <div id="About" class="tab">
203             <h1 class="text-2xl font-bold mb-4 flex items-center">
204                 <i data-lucide="smartphone-nfc" class="w-6 h-6 text-blue-500 mr-2"></i>
205                 What is electromagnetic radiation?
206             </h1>
207             <div class="bg-white p-4 rounded-lg shadow-md flex items-center justify-between about-card">
208                 <div>
209                     <p class="text-sm text-gray-500">Electromagnetic radiation, in short EMR, is a se
t of waves which distributes through space, without a physical conductor. It consists of oscillating elec
tric and magnetic fields that travel at the speed of light.
210                 </p>
211                 </div>
212             </div>
213             <h2 class="text-xl font-semibold mt-6"></h2>
214             <h1 class="text-2xl font-bold mb-4 flex items-center">
215                 <i data-lucide="microchip" class="w-6 h-6 text-blue-500 mr-2"></i>
216                 EMR-Mapping Device:
217             </h1>
218             <div class="bg-white p-4 rounded-lg shadow-md">
219                 <p class="text-sm text-gray-500">
220                     The EMR-Mapping Device is a portable tool designed to measure electromagnetic radiation (EMR) lev
els in the environment. It collects real-time data on EMR intensity and records it alongside GPS coordina
tes, allowing for the creation of detailed radiation maps. The device features a built-in display for imm
ediate feedback and stores the collected data for further analysis. With its ability to track EMR levels
across different locations, the device provides valuable insights into electromagnetic exposure in our su
rroundings.
221                 </p>
222             </div>

```

```

224 <h2 class="text-xl font-semibold mt-6"></h2>
225
226 <div class="grid grid-cols-1 md:grid-cols-3 gap-6">
227   <div class="bg-white p-4 rounded-lg shadow-md">
228     
229   </div>
230   <div class="bg-white p-4 rounded-lg shadow-md">
231     
232   </div>
233   <div class="bg-white p-4 rounded-lg shadow-md">
234     
235   </div>
236 </div>
237   <h2 class="text-xl font-semibold mt-6"></h2>
238   <h1 class="text-2xl font-bold mb-4">Team</h1>
239   <div class="team-container">
240     <div class="card">
241       
242       <div class="details">
243         <label>Lisa Kolli</label>
244         <p>Made the Hardware part of our Diploma<br>
245           <i>Hardware, Team Leader</i>
246         </p>
247       </div>
248     </div>
249     <div class="card">
250       
251       <div class="details">
252         <label>Letissa Adlboller</label>
253         <p>Made the Literature part of our Diploma<br>
254           <i>Literature Research, Team Member</i>
255         </p>
256       </div>
257     </div>
258     <div class="card">
259       
260       <div class="details">
261         <label>Robin Heindl</label>
262         <p>Made the software part of our Diploma<br>
263           <i>Software, Team Member</i>
264         </p>
265       </div>
266     </div>

```

```

269 </div>
270 </div>
271
272 </div>
273 <div id="map" class="tab hidden">
274   <h1 class="text-2xl font-bold mb-4">Heatmap</h1>
275   <div class="bg-white rounded-lg shadow-md p-4">{map_html}</div>
276 </div>
277 <div id="stats" class="tab hidden">
278   <h1 class="text-2xl font-bold mb-4">Statistics</h1>
279   <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
280     <div class="bg-white p-4 rounded-lg shadow-md flex items-center justify-between stat-card">
281       <div>
282         <p class="text-sm text-gray-500">Average EMR</p>
283         <p class="text-xl font-bold text-gray-800">{average_emf:.2f} mV</p>
284       </div>
285       <i data-lucide="radio-tower" class="w-6 h-6 text-blue-500"></i>
286     </div>
287     <div class="bg-white p-4 rounded-lg shadow-md flex items-center justify-between stat-card">
288       <div>
289         <p class="text-sm text-gray-500">Max EMR</p>
290         <p class="text-xl font-bold text-gray-800">{max_emf:.2f} mV</p>
291       </div>
292       <i data-lucide="arrow-up" class="w-6 h-6 text-green-500"></i>
293     </div>
294
295     <div class="bg-white p-4 rounded-lg shadow-md flex items-center justify-between stat-card">
296       <div>
297         <p class="text-sm text-gray-500">Min EMR</p>
298         <p class="text-xl font-bold text-gray-800">{min_emf:.2f} mV</p>
299       </div>
300       <i data-lucide="arrow-down" class="w-6 h-6 text-red-500"></i>
301     </div>
302
303     <div class="bg-white p-4 rounded-lg shadow-md flex items-center justify-between stat-card">
304       <div>
305         <p class="text-sm text-gray-500">Standard Devision</p>
306         <p class="text-xl font-bold text-gray-800">{std_deviation:.2f} mV</p>
307       </div>
308       <span style="font-size: 1.7em; color: #FFD700;">&#963;</span>
309     </div>
310     <div class="bg-white p-4 rounded-lg shadow-md flex items-center justify-between stat-card">
311       <div>
312         <p class="text-sm text-gray-500">Count</p>
313         <p class="text-xl font-bold text-gray-800">{count_emf}</p>
314       </div>
315       <i data-lucide="chart-no-axes-column-increasing" class="w-6 h-6 text-indigo-500"></i>
316     </div>
317     <div class="bg-white p-4 rounded-lg shadow-md flex items-center justify-between stat-card">
318       <div>
319         <p class="text-sm text-gray-500">Range</p>
320         <p class="text-xl font-bold text-gray-800">{range_emf:.2f} mV</p>
321       </div>
322       <i data-lucide="move-horizontal" class="w-6 h-6 text-purple-500"></i>
323     </div>

```

```

327         </div>
328         <h2 class="text-xl font-semibold mt-6"></h2>
329         <h1 class="text-2xl font-bold mb-4">Charts</h1>
330         <h2 class="text-xl font-semibold mt-6"></h2>
331         
332         <h2 class="text-xl font-semibold mt-6"></h2>
333         
334         <h2 class="text-xl font-semibold mt-6"></h2>
335         
336     </div>
337 </main>
338 <script>
339     lucide.createIcons();
340     function showTab(tab) {{
341         document.querySelectorAll('.tab').forEach(div => div.classList.add('hidden'));
342         document.getElementById(tab).classList.remove('hidden');
343         document.querySelectorAll('button').forEach(btn => btn.classList.remove('bg-blue-500', 'text-white'));
344         document.getElementById(tab + 'Btn').classList.add('bg-blue-500', 'text-white');
345     }}
346 </script>
347 </body>
348 </html>
349 ""
350
351 # HTML speichern
352 with open(html_output_path, "w", encoding="utf-8") as f:
353     f.write(html_template)
354
355 print(f"Dashboard wurde erstellt: {html_output_path}")

```

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